

Sensitivity to Subjective Expected Utility Maximization: A Methodological Study, with an Illustrative Application to LLM Decision-Making

Working draft (pre-v1) — comments welcome

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Abstract

Evaluating the quality of decisions made under uncertainty is hard when labeled outcomes are scarce, costly, or confounded with luck. We treat subjective expected utility (SEU) maximization as a *stated standard* and define a graded measure — *SEU sensitivity* — of an agent’s conformity to it. The formal vehicle is a softmax choice model with a sensitivity parameter α applied to SEU-valued alternatives; the methodological contribution is a sequence of identifiability results for α and for the belief and utility parameters (β, δ) , validated in Stan via prior predictive checks, parameter recovery, and simulation-based calibration (SBC). The findings are reported with their finite-sample caveats intact. In the uncertain-choice-only model $\mathbf{m}_0$, α is identifiable but (β, δ) are only weakly informed by uncertain choices: the joint posterior barely contracts and concentrates on a β – δ trade-off at realistic sample sizes. In the extended model $\mathbf{m}_1$, δ becomes identifiable in principle (hence estimable in the limit) via a β -free risky block, but its practical recovery gain at realistic sample sizes is negligible (a matched-count CI-width reduction under 1%); and at matched choice count the β -free risky block yields no α -precision gain either, so finite-sample α precision is driven by data *quantity*, not the *type* of choice. These are two distinct phenomena: for δ , identifiability does not imply precise estimability at realistic n ; and for α , identifiability in principle does not by itself reveal what governs its precision at finite n — which we find to be the *quantity* of choice data, not the *type* of choice. Marginal SBC passes for both models even where the joint (belief, utility) posterior is only weakly informed — a *marginal-SBC demarcation* we make precise. A two-by-two illustrative application (GPT-4o and Claude 3.5 Sonnet, each on insurance-claims triage and Ellsberg-style urns, with sampling temperature as the lever) shows the workflow runs end-to-end on real LLM choice data, detecting a structured comparative α effect in two of four cells and declining to support one in the other two.

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Motivation

1.1 Two approaches to assessing a decision maker

There are two broad approaches to assessing a decision maker from a body of choices. The first is *external*, or label-based: do the agent’s choices agree with a labeled set of decisions held to be correct — a key that records, for each decision problem, which alternative an informed judge would select; would such a judge have forwarded the same claim, taken the same gamble, made the same call? The criterion is agreement with a curated answer key, whoever supplies it. The second is *internal*: are the choices coherent with one another in the way a stated standard of consistency requires — responsive to the probabilities, the stakes, and the tradeoffs in the way the standard prescribes? We assess this second kind of conformity from the agent’s choices as observed, without attributing to the agent any particular internal computation or deliberative procedure: “internal” qualifies the *standard* — coherence among the agent’s own choices — not the agent’s cognition. The standard constrains behavior, and the question is whether the behavior conforms to it.

The two assessments answer to different things and can come apart. The consistency standard constrains how an agent’s choices hang together, not which alternative an external key marks as correct: an agent whose beliefs and values differ from those the key encodes can be fully consistent yet disagree with it, and an agent can match the key on a given decision while its choices violate the standard elsewhere. The internal-consistency assessment is the one this paper develops, and it is especially relevant where a correct-choice key is unavailable or costly to construct — in assessing a human analyst, an institution, or, increasingly, a large language model deployed to make or recommend choices.

Both approaches have a long precedent in the study of decision makers. One strand of internal-consistency critique assesses choices against expected utility theory as a standard for decision making under risk: the Allais paradox (Allais 1953) exhibits a pattern of preferences that violates the independence axiom, a tension Kahneman and Tversky took up in developing prospect theory (Kahneman and Tversky 1979; Tversky and Kahneman 1992). These demonstrations, and the large-scale replication that subsequently confirmed their patterns (Ruggeri et al. 2020), were built on *pairwise* choices — a restriction well suited to their purpose of isolating axiom violations — whereas the instrument developed here places no such restriction on the number of alternatives in a decision problem, which opens the possibility of probing how sensitivity itself varies with menu size (an extension we return to in §8.5). A second strand — the heuristics-and-biases program — assesses judgments about uncertainty against the probability axioms, construed as a coherence standard on degrees of belief in the Dutch-book tradition (Ramsey 1926; Finetti 1937): the conjunction fallacy of the “Linda problem” (Tversky and Kahneman 1983) exhibits credences that violate that calculus, evidence used to motivate the representativeness heuristic. In each case the decision maker

is faulted not against a labeled key but against a coherence requirement that its own choices or judgments fail to meet. The contrasting tradition scores choices against an externally curated set of good options: the choice architecture of Thaler and Sunstein (2008), for instance, assesses diners against a menu whose “healthy” items have been labeled by dieticians. The instrument developed here belongs to the internal-consistency tradition — it measures conformity to an internal standard — with subjective expected utility maximization, which couples a coherence requirement on degrees of belief with one on preferences, in the role the axioms of rational choice play in those precedents.

1.2 Challenges for label-based assessment of decision quality

The label-based approach scores choices against a key, drawn either from an adjudicated set of correct choices or from the realized outcomes of the decisions. Two difficulties limit what such a key can tell us about the quality of the decisions: a key of correct choices is often impractical to obtain, and a key of outcomes measures the wrong thing.

The first concerns obtaining the labels at all. A key of correct choices records, decision by decision, which alternative is the right one to select, and assembling one is often impractical. The insurance-claims study we report in §7 is a case in point: there is no off-the-shelf key that says which choice was correct, and constructing one would mean asking subject-matter experts what they would choose across the decision problems — an exercise that is costly and, because expert judgment varies, sensitive to which experts are consulted. We could not produce such a key credibly under realistic experimental conditions.

The second difficulty is more basic, and remains even where a key *is* at hand — in particular where it records realized *outcomes* rather than adjudicated choices. A good outcome is not the same as a good decision: the outcome of a choice typically depends on how the relevant uncertainty happens to resolve, which lies outside the decision maker’s control. A score computed purely from outcomes also ignores the way the context in which a decision is made informs the beliefs and desires of the decision maker. Treating outcomes as a direct measure of decision quality is defensible only under strong assumptions — for instance, that the decision maker is properly incentivized and has enough information to learn the “objective” probabilities of the relevant states. Those assumptions are reasonable in some controlled settings, but we would not want to impose them in a study like the insurance experiment of §7, where neither condition can be taken for granted. A decision that is well structured given the agent’s information may still be scored poorly by an outcome key that was generated under different information, and vice versa.

1.3 The need for consistency-based evaluation

What remains available when neither kind of key can be had is the *structure* of the choices themselves. If we are willing to state a standard of good decision making explicitly, we can ask whether an agent’s choices are responsive to probabilities, utilities, and tradeoffs in the way that standard prescribes — without a labeled set of correct choices, and without reference to which outcome happened to be realized. Evaluation of this internal-consistency kind replaces a claim about outcomes with a claim about conformity to a stated standard. It does not require any labeled set of decisions; it requires only that the standard be named explicitly. In return it yields an evaluation that is well defined even where no key of correct choices and no outcome record exist, and that does not depend on which outcome happened to be realized.

1.4 SEU as the stated standard

We take *subjective expected utility (SEU) maximization* as the reference standard. An SEU-committed agent holds a subjective probability over the states of the world and a utility over consequences, and prefers the alternative whose expected utility is greatest (Savage 1954a; Anscombe and Aumann 1963; Neumann and Morgenstern 1947). We do *not* argue that SEU is the uniquely correct theory of rational choice, and nothing below depends on that stronger claim. The argument is conditional: *if* SEU is the reference standard, then a graded, statistically rigorous measure of conformity to it is methodologically useful. SEU is the natural first case for such an instrument: it is the most widely shared normative benchmark in decision theory, so a conformity measure built on it reaches the largest audience. The identifiability questions that occupy §§3 and 5 are then taken up for that instrument in turn.

Two qualifications bound the sense in which other standards could be substituted for SEU. First, several theories often mentioned alongside SEU are not normative standards of decision quality at all but descriptive accounts of how people actually choose; cumulative prospect theory (Kahneman and Tversky 1979; Tversky and Kahneman 1992) is the clearest example. Substituting such a theory is therefore more than a change of functional form: the quantity being measured changes from conformity to a norm to fit to a behavioral model, and the interpretation of the instrument changes with it. Maxmin expected utility (Gilboa and Schmeidler 1989) is more plausibly read as a normative alternative, and it supplies an admissible value function for the softmax construction; but because that value function is a *minimum* over a set of priors, it is not smooth, and substituting it into the gradient-based (Hamiltonian Monte Carlo) implementation of §§4 and 6 is not a drop-in replacement — the kinks of the min would require either a smoothing that reintroduces a precision parameter confoundable with α , or a non-gradient sampler. Smooth ambiguity models (Klibanoff, Marinacci, and Mukerji 2005) avoid the non-smoothness by construction and would substitute more readily. Cumulative prospect theory, by contrast, could not be substituted at all without abandoning the normative reading. Second, the template applies only to standards that rank alternatives by a single real-valued evaluation. Standards that decline to reduce choice to such an evaluation — for instance the decision theory of Levi (1980, 1986), in which admissible options need not be totally ordered — fall outside the framework entirely, not merely outside its current scope. Parallel instruments for the real-valued non-SEU standards are possible by the same template and are out of scope here.

1.5 Why a graded measure rather than a binary verdict

Real decision makers, human or machine, only ever *approximately* satisfy any rationality axiom (Simon 1955; Lieder and Griffiths 2020). A binary verdict — “rational” or “not” — discards exactly the information that an evaluation should preserve, namely *how closely* and *how reliably* an agent’s choices track the standard. We therefore measure conformity on a continuum. The formal vehicle, developed in §2, is a softmax (Luce–McFadden) choice model (Luce 1959; McFadden 1974; Train 2009) carrying a single scalar *sensitivity* parameter $\alpha \geq 0$ applied to SEU-valued alternatives. As α ranges from 0 to ∞ , the implied behavior runs from uniform-random choice (insensitive to expected utility) to deterministic SEU maximization (perfectly sensitive to it). The empirically interesting agents live strictly between these poles, and α is the coordinate that locates them there. We call α the agent’s **SEU sensitivity**: the disposition of an agent committed to SEU maximization to act in accordance with that commitment, held separate from both the *content* of the commitment (the agent’s beliefs and utilities) and the *outcomes* it happens to realize.

1.6 Scope

This paper is methodological. Its focus is a calibrated measurement instrument and the identifiability analysis that licenses it; the decision-theoretic and philosophical material frames that analysis. §§2–6 specify, motivate, and validate the instrument: the abstract choice model and its three characterizing properties (§2), the identifiability of α together with the finding that the belief and utility parameters (β, δ) are only *weakly informed* in the finite-sample Bayesian workflow for an uncertain-choice-only model (§3), a Stan implementation validated by prior predictive checks, parameter recovery, and simulation-based calibration (SBC) (§4), and the same arc for an extended model that adds risky choices (§§5–6). §7 then runs the full workflow end-to-end on real LLM choice data: a 2×2 illustrative application crossing GPT-4o and Claude 3.5 Sonnet with insurance-claims triage and Ellsberg-style urn gambles (Ellsberg 1961), using sampling temperature as the experimental lever. The application is included to show that the methodology does real evaluative work on real choice data (Binz and Schulz 2023; Hagendorff, Fabi, and Kosinski 2023); it is *not* a substantive contribution to LLM behavioral science, and §7.6 draws that line explicitly.

Several things are deliberately out of scope, and are flagged once at their natural locations rather than revisited: an information-optimal lottery design for sharpening the utility parameter (future work, §8.5); the hierarchical extension `h_m01` and the multi-LLM alignment study it was built for (a planned companion paper, §7.7); generalized-sensitivity models with block-specific or context-specific α ; a test for ambiguity aversion in the Ellsberg setting (the SEU-plus-softmax model is, by assumption, of the wrong form for that, §7.6); and any absolute-rationality ranking *between* LLMs (only within-design comparative claims about α are licensed).

1.7 What is new

Multinomial logit is McFadden’s (McFadden 1974); Stan, SBC, and the prior-predictive/recovery/SBC workflow have been standard practice for a decade (Carpenter et al. 2017; Talts et al. 2018; Gelman et al. 2020). The contribution here is not any one of these tools. It is, first, a *conceptual* contribution — item (a) below: a principled explication of what it means for a decision maker to be *sensitive to*, or *aligned with*, the requirements of SEU maximization across a range of decision contexts — and, resting on that explication, a four-part *methodological* package, items (b)–(e), that makes the explication measurable and validates the measurement.

(a) A principled explication of SEU sensitivity. Within the softmax/Luce–McFadden specification, three characterizing properties of the choice rule (§2.3, proved in Appendix A) license reading its single scalar parameter α as the degree to which an agent’s choices track SEU maximization, with $\alpha = 0$ at indifference to expected utility and $\alpha \rightarrow \infty$ at deterministic maximization. This gives a graded, well-defined meaning to “alignment with expected-utility reasoning” across the family of decision problems the estimable model accommodates (the experiments expressible in the data block of the Stan models, e.g., `m_01`), and separates that disposition from the *content* of the agent’s commitments (its beliefs and utilities) and from the *outcomes* it realizes (§2.5). The generalization across many agents and contexts via a hierarchical model is left to a companion paper (§7.7). *In plain terms:* α assigns a precise number to how closely an agent’s choices follow expected-utility reasoning, holding apart what the agent believes, what it wants, and how things happen to turn out.

(b) A clean identifiability decomposition. We separate the identifiability of α *from the expected-utility vector* η — stated as a proposition under an explicit genericity condition (§3.3, Appendix B.1) — from the question of whether the data also pin down the ingredients of η , namely beliefs (β) and utilities (δ) . At realistic sample sizes they do not: uncertain choices leave (β, δ) only *weakly*

informed, as the Bayesian workflow shows directly — wide, correlated (β, δ) posteriors against tight α intervals (§3.4, §4.3). The two are one structural picture (§3.5): given the ν -endpoint convention that fixes the utility scale, α is read off the curvature of choice probabilities in η , while beliefs and utilities enter only *through* η and trade off against each other in a way that uncertain-choice data barely resolve. *In plain terms*: the alignment number α can be recovered from choice data, but the agent’s underlying beliefs and utilities often cannot be told apart from one another by that same data.

(c) A worked demonstration that in-principle identifiability does not predict finite-sample estimability. The m_0/m_1 contrast makes the gap concrete in two distinct ways. For the utility parameter δ , adding the risky block makes it identifiable in principle (§5.5) yet buys almost no precision at the design sample size — a matched-count credible-interval-width reduction under 1% (§6.4). For the sensitivity parameter α , a matched recovery study shows that finite-sample precision is governed by data *quantity*, not by the *type* of choice that an in-principle argument might privilege. *In plain terms*: a quantity that is recoverable in theory may still not be recoverable from a realistic amount of data.

(d) The marginal-SBC demarcation. Marginal rank uniformity is *necessary but not sufficient* for joint posterior calibration. Separately, the correlated, barely-contracted joint (β, δ) posterior — which can itself be perfectly calibrated — reflects weak joint *informativeness*, and marginal SBC does not reveal it: per-parameter ranks pass for both models even where beliefs and utilities are jointly only weakly resolved (§§4.4, 6.5, 8.3). *In plain terms*: a standard calibration check can look healthy one parameter at a time while still concealing that two parameters are jointly unresolved.

(e) An illustrative full-pipeline application of the instrument to LLM choice data, in which the α inference and the posterior-predictive summaries come in clean — the only flagged diagnostics being confined to the weakly-informed nuisance parameters (§7.4) — and the framework reports posterior support for a structured comparative effect in two of four LLM $\times$ task cells while withholding it in the other two, where the design’s resolution is insufficient (§7). *In plain terms*: run end-to-end on real language-model choices, the instrument’s sensitivity estimates behave as designed and report an effect in two of four settings, with the non-detections reflecting limited resolution rather than established absence.

1.8 Roadmap

The paper alternates between *abstract/identifiability* sections and their *computational* realizations, then applies the result. §2 fixes the abstract model. §3 establishes the identifiability picture for the uncertain-choice-only model m_0 , and §4 realizes and validates it in Stan. §5 extends the model with risky choices and revisits identifiability, and §6 realizes and validates the extension — inheriting both the design and the identifiability questions from the earlier pair. §7 runs the validated workflow on real LLM data, and §8 draws the methodological lessons together. The formal core — the proofs that close §§3 and 5 — is in Appendix B; the body states results and points to it. Concretely, the dependency structure is

$$\underbrace{\S 2}_{\text{model}} \longrightarrow \underbrace{\S 3 \rightarrow \S 4}_{m_0} \longrightarrow \underbrace{\S 5 \rightarrow \S 6}_{m_1} \longrightarrow \underbrace{\S 7}_{\text{application}} \longrightarrow \underbrace{\S 8}_{\text{discussion}},$$

with §3 feeding *both* §4 and §5, the two implementation sections (§§4, 6) converging on the application, and Appendix B supplying the proofs that close the two identifiability sections.

The Abstract Model: Softmax Choice and Sensitivity

This section specifies the choice model abstractly, before any commitment to where values come from. The payoff of the abstraction is that the three properties that license reading α as a *sensitivity* parameter (§2.3) hold for *any* value function; the subjective expected utility (SEU) specialization of §2.4 then inherits them as corollaries.

2.1 Alternatives, values, and the sensitivity parameter

Let $\mathcal{R} = \{1, \dots, R\}$ be a finite set of distinct *alternatives*. A *value function* $V : \mathcal{R} \rightarrow \mathbb{R}$ assigns a real number $V(r)$ to each alternative. A decision problem m presents a subset of alternatives, encoded by availability indicators $I_{m,r} \in \{0, 1\}$. A *sensitivity parameter* $\alpha \geq 0$ governs how sharply choices track value differences.

2.2 The softmax choice rule

The probability of selecting alternative r from the available set in problem m is the **softmax** (Luce–McFadden–Boltzmann) rule

$$P(\text{choose } r \mid \alpha, V) = \frac{\exp(\alpha V(r))}{\sum_{j: I_{m,j}=1} \exp(\alpha V(j))}. \quad (1)$$

The functional form has independent precedents: Luce’s (1959) ratio-scale choice axiom, McFadden’s (1974) random-utility derivation of multinomial logit, and the Boltzmann distribution of statistical mechanics (where α is inverse temperature). We use α rather than a temperature $T = 1/\alpha$ because *higher* α corresponds to *higher* sensitivity, the more intuitive direction for our purposes.

Why softmax rather than probit or another stochastic-choice wrapping? Three features recommend it. Luce’s (1959) independence-of-irrelevant-alternatives axiom yields a ratio form $P(r \mid \mathcal{A}) = w(r) / \sum_{j \in \mathcal{A}} w(j)$ over positive weights; specifying an exponential link $w(r) = \exp(\alpha V(r))$ from the real-valued value scale to those weights — equivalently, log-odds linearity in value differences — selects the softmax (Equation 1). Given that link, softmax embeds the optimization and uniform-choice limits as the two endpoints of a single scalar α (Properties 2 and 3 of §2.3, proved as Theorems A.2 and A.3 in Appendix A), and its log-likelihood is concave in the index αV — a standard multinomial-logit property (McFadden 1974; Train 2009) — so that, with the value scale V held fixed, the α -likelihood is unimodal and free of spurious local optima. (The concavity is in the index αV , i.e. the surface in α with V fixed; it is *not* joint concavity in α together with the value parameters, because αV is bilinear in the two.) These are two distinct virtues, and we keep them apart: the log-odds linearity and single-scalar limit structure above make α *interpretable*, while the concavity is what makes it *well-behaved to estimate*. Probit, by contrast, replaces the Luce ratio with Gaussian latent errors and so offers neither a closed-form IIA/log-odds structure nor a single sensitivity scalar carrying these two limit properties (Train 2009).

2.3 Three characterizing properties of softmax

The following hold for *any* value function V , within each decision problem m and relative to its available set $\mathcal{A}_m = \{r : I_{m,r} = 1\}$. Write $V_m^* = \max_{r \in \mathcal{A}_m} V(r)$ for the maximal available value, $\mathcal{R}_m^* = \{r \in \mathcal{A}_m : V(r) = V_m^*\}$ for the value-maximizing set, and $\mathcal{R}_m^- = \mathcal{A}_m \setminus \mathcal{R}_m^*$. Proofs are in Appendix A.

i Property 1 (Monotonicity in sensitivity). Holding V fixed, for any two available alternatives $r, s \in \mathcal{A}_m$ the log-odds are linear in α ,

$$\frac{\partial}{\partial \alpha} \log \frac{P(\text{choose } r \mid \alpha, V)}{P(\text{choose } s \mid \alpha, V)} = V(r) - V(s),$$

so the odds always shift monotonically toward the higher-valued alternative as α increases. Consequently the *total* probability on the value-maximizing set $\mathcal{R}_m^*$ is strictly increasing in α ; in particular a single value-maximizing alternative ($r \in \mathcal{R}_m^*$) gains and a value-minimizing alternative loses probability. (An individual non-maximal alternative whose value exceeds the current choice-weighted mean can rise before it falls, so non-maximal probabilities are not monotone one-by-one.)

i Property 2 (Optimization limit). As $\alpha \rightarrow \infty$, choice concentrates on the value-maximizing set: $P(\text{choose } r \mid \alpha, V) \rightarrow 1/|\mathcal{R}^*|$ for $r \in \mathcal{R}^*$ and $\rightarrow 0$ otherwise. With a unique maximizer (the generic case under continuous priors) choice becomes deterministic.

i Property 3 (Uniform-choice limit). As $\alpha \rightarrow 0$, choice approaches the uniform distribution over available alternatives, independent of V .

Together the three properties give exactly the conceptual content needed to read α as “sensitivity to value maximization,” with no reference yet to where values come from: α interpolates monotonically (Property 1) between indifference to value (Property 3) and exclusive pursuit of the maximum (Property 2).

2.4 SEU specialization

Specialize the value function to a *subjective expected utility*. Fix K consequences with an (ordered) utility vector $\mathbf{v} \in \mathbb{R}^K$. Each alternative r carries a subjective probability distribution $\boldsymbol{\psi}_r \in \Delta^{K-1}$ over consequences, and its value is its expected utility

$$V(r) = \eta_r = \boldsymbol{\psi}_r^\top \mathbf{v} = \sum_{k=1}^K \psi_{r,k} v_k. \quad (2)$$

Because η_r is just a particular value function, Properties 1–3 apply verbatim, and α now measures sensitivity to *SEU maximization* specifically: the disposition to choose the expected-utility-maximizing alternative.

2.5 The conceptual payoff

Sensitivity to SEU maximization is the disposition of an SEU-committed agent to *act in accordance with its commitments*. The contrast between an agent’s *commitment* to a standard and its *performance* relative to that standard is Levi’s (Levi 1980, chap. 1); we return to it, and to the sense in which α measures a tendency to perform, in §8.1. The disposition is distinct from (a) the *content* of those commitments — the agent’s beliefs $\boldsymbol{\psi}$ and utilities $\mathbf{v}$ — and (b) the *outcomes* the agent happens to realize. The SEU construction thus decomposes choice behavior conceptually into beliefs, utilities, and sensitivity. The three pieces are *conceptually* separable in this way, but they

are not equally separable *empirically*: as §3.4 and §5.5 show, α separates cleanly from the rest, whereas β and δ are only weakly informed by uncertain choices and are recovered jointly rather than each in isolation.

⚠ Caveat on uniqueness of the decomposition. The (α, ψ, v) decomposition is not the only way to organize the same observable choice data: rank-dependent utility, cumulative prospect theory (Tversky and Kahneman 1992), and maxmin expected utility (Gilboa and Schmeidler 1989) organize it differently. These alternatives are not interchangeable substitutes for SEU, and they differ among themselves in kind — some are descriptive accounts rather than normative standards, as §1.4 discusses. Our claim is only that *under* an SEU reference standard the (sensitivity, beliefs, utilities) decomposition is well-defined and — with the identifiability caveats of §3 and §5 — measurable. It is not a claim that this decomposition is uniquely correct for all theoretical purposes.

2.6 Notation, the utility-scale convention, and the β gauge

For the parameterized models of §3 and §5 we generate subjective probabilities from D -dimensional feature vectors w_r via a linear-softmax belief map $\psi_r = \text{softmax}(\beta w_r)$ with $\beta \in \mathbb{R}^{K \times D}$, in the multinomial-logit tradition (McFadden 1974; Train 2009). The utility vector is built from increments $\delta \in \Delta^{K-2}$ via cumulative sums, with endpoints fixed by convention. (The $K - 1$ increments $\delta_j = v_{j+1} - v_j$ are nonnegative and sum to $v_K - v_1 = 1$, so δ lies in the $(K - 2)$ -simplex Δ^{K-2} , matching the Stan declaration `simplex[K - 1] delta`.)

Two facts about this parameterization recur throughout and are stated once here, then referenced rather than redefined. The first fixes the zero and unit of the utility scale, which is what makes α a well-posed target; the second records an indeterminacy of the belief map.

! The v -endpoint convention. vNM utilities are unique only up to a positive affine transformation $\tilde{v}_k = a v_k + b$ with $a > 0$, and rescaling by a is observationally equivalent to rescaling α by $1/a$ (Theorem A.4). Fixing the endpoints

$$0 = v_1 \leq v_2 \leq \cdots \leq v_K = 1$$

fixes the zero and unit of the utility scale, removing this affine indeterminacy and making α identifiable as sensitivity on a standardized one-unit utility scale.

! The β gauge (additive row-shift). Adding the same vector $\gamma \in \mathbb{R}^D$ to every row of β — that is, $\beta_k \mapsto \beta_k + \gamma^\top$ for all k — shifts every entry $(\beta w_r)_k$ by the common amount $\gamma^\top w_r$ and so leaves $\text{softmax}(\beta w_r)$ unchanged. We rely on no other invariance of the belief map; we refer to it as the β **gauge** and reference it in §3.3, §4, §6, and Appendix B.4.

The consolidated glossary is collected once here (Table 1) and referenced, not redefined, elsewhere.

Table 1: Consolidated notation.

Symbol	Meaning	First used
$\alpha \geq 0$	sensitivity to value maximization (primary quantity of interest)	§2.1
$V(r), \eta_r$	value / expected utility of r , $\eta_r = \boldsymbol{\psi}_r^\top \boldsymbol{v}$	§2.1, §2.4
$\boldsymbol{\psi}_r$	subjective probabilities for uncertain alternative r (from β)	§2.4
$\beta \in \mathbb{R}^{K \times D}$	belief-formation weights; $\boldsymbol{\psi}_r = \text{softmax}(\beta w_r)$	§2.6
w_r	D -dimensional feature vector of alternative r	§2.6
$\delta \in \Delta^{K-2}$	utility increments ($K - 1$ entries; Stan <code>simplex[K - 1]</code>)	§2.6
$\boldsymbol{v}$	ordered utilities, $v_1 = 0 \leq \dots \leq v_K = 1$	§2.6
π_s	objective lottery simplex for risky alternative s (Stan: <code>x_risky</code>)	§5.2
β gauge	additive row-shift indeterminacy of the belief softmax	§2.6
$\boldsymbol{v}$ -endpoint convention	fixes the zero and unit of the vNM utility scale	§2.6

2.7 The identifiability question

The choice-probability function is determined by $(\alpha, \boldsymbol{\psi}, \boldsymbol{v})$. Whether the *parameterizations* of $\boldsymbol{\psi}$ and $\boldsymbol{v}$ — namely (β, δ) — are recoverable from observed choices is the question §3 and §5 take up.

Choice Under Uncertainty Alone: Identifiability of α , and the Weakly-Informed (β, δ)

3.1 The parameterization

Recall from §2.6 the uncertain-choice (“m_0”) model. Subjective probabilities are generated from D -dimensional features w_r via $\boldsymbol{\psi}_r = \text{softmax}(\beta w_r)$ with $\beta \in \mathbb{R}^{K \times D}$. Utilities are constructed from increments $\delta \in \Delta^{K-2}$ (the $K - 1$ nonnegative increments summing to one; Stan `simplex[K - 1] delta`) by cumulative summation, yielding $0 = v_1 \leq v_2 \leq \dots \leq v_K = 1$. Expected utilities are $\eta_r = \boldsymbol{\psi}_r^\top \boldsymbol{v}$, and choices follow Equation 1 with $V(r) = \eta_r$. The β gauge (§2.6) is the only β indeterminacy we use.

3.2 Why this parameterization

Linear-softmax for ψ aligns with the multinomial-logit / discrete-choice tradition (McFadden 1974; Train 2009); ordered utilities with endpoints fixed by convention remove the affine scale-and-shift indeterminacy of vNM utility (Theorem A.4). The two conventions together leave α as a scalar with the limit interpretation of §2.3.

3.3 Identifiability of α from η

i Proposition 3.1 (Identifiability of α from η). Fix the expected-utility vector $\eta = (\eta_r)_{r \in \mathcal{R}}$. If there is a menu $\mathcal{M} \subseteq \mathcal{R}$ of positive design probability on which the values $\{\eta_r : r \in \mathcal{M}\}$ are not all equal, then $\alpha > 0$ is globally identifiable from the choice-probability function on $\mathcal{M}$.

Sketch (full proof in Appendix B.1). On any menu with non-constant η the log-odds between two alternatives with $\eta_r \neq \eta_s$ is $\log[P(r)/P(s)] = \alpha(\eta_r - \eta_s)$, strictly monotone in α ; the softmax choice-probability map is therefore injective in α on such a menu.

i Boundary remark ($\alpha = 0$). Proposition 3.1 identifies α throughout the open cone $\alpha > 0$, and the log-odds identity $\log[P(r)/P(s)] = \alpha(\eta_r - \eta_s)$ is injective on the closed half-line $[0, \infty)$ whenever $\eta_r \neq \eta_s$, so $\alpha = 0$ is reached continuously from above and the genericity condition on $\{\eta_r\}$ is *not* vacuous there. With η fixed the softmax log-likelihood $\ell = \alpha\eta_y - \log \sum_j \exp(\alpha\eta_j)$ is in fact smooth at $\alpha = 0$, with $\partial_\alpha \ell|_{\alpha=0} = \eta_y - \bar{\eta}$. What is special about $\alpha = 0$ is therefore not non-differentiability but *non-regularity*: the choice rule is uniform (Theorem A.3) and derivatives with respect to any parameter entering only through η are multiplied by α , so β and δ become locally unidentified at the boundary. The proposition does not silently exclude the near-random null; it simply does not assert differentiable identification of the η -generating parameters *at* the boundary.

The structure of the claim matters: α is identified *from* η , not from (β, δ) directly. Whether (β, δ) themselves are recoverable from uncertain choices is a separate question — the subject of §3.4.

3.4 Why (β, δ) are weakly informed by uncertain choices

Proposition 3.1 identifies α *from* η . The remaining question is whether the data also pin down the two ingredients that *make up* η — beliefs (β) and utilities (δ) . They do not, at realistic sample sizes, and the reason is structural. Expected utility composes the two multiplicatively,

$$\eta_r = \text{softmax}(\beta w_r)^\top \mathbf{v}(\delta),$$

the product of a β -driven simplex and a δ -driven utility vector. Uncertain choices see (β, δ) only through this scalar η_r , so a family of (β, δ) pairs that trades a change in beliefs against a compensating change in utilities leaves the implied expected utilities — and hence the choice probabilities — almost unchanged. The $\mathbf{v}$ -endpoint convention (§2.6) fixes the zero and unit of the utility scale, but it does not touch this *multiplicative* (β, δ) coupling, which is not an indeterminacy that any convention removes; the coupling is what leaves β and δ only weakly informed by uncertain-choice data.

This shows up directly in the Bayesian workflow rather than as a separate formal claim. Parameter recovery in `m_0` (§4.3) returns wide marginal credible intervals for both β and δ — in contrast to

the tight, well-calibrated α intervals — together with *correlated β - δ estimation errors*: concretely, across recovery replicates the posterior-mean estimation errors of a representative weight entry $\beta_{1,1}$ and a representative increment δ_1 are correlated (Pearson; §4.3). This is an *across-replicate* error correlation, not a within-posterior one; because raw β entries carry the row-shift gauge we read this representative-component diagnostic as an illustrative signature of the coupling rather than a gauge-invariant statistic, and we do not attach significance to its sign (which varies with the true- α regime). The recovered posteriors concentrate not on (β, δ) individually but on a compensating trade-off between them. That correlated, barely-contracted joint posterior is the practical signature of the multiplicative coupling, and it is what we mean throughout by calling (β, δ) *weakly informed* by uncertain choices.

We state this as an empirical feature of the posterior, not as a theorem: no strict non-identifiability is claimed, and no gauge group on (β, δ) beyond the β row-shift of §2.6 is asserted. The point is that uncertain-choice data alone supply little information for separating beliefs from utilities — which is exactly what motivates the extended model of §5.

3.5 The two facts are one structural picture

α is identified *from* η (Proposition 3.1), whereas the two ingredients of η — beliefs and utilities — are only weakly informed by uncertain choices (§3.4). These are two sides of one fact: the map $(\beta, \delta) \rightarrow \eta$ collapses beliefs and utilities multiplicatively into a single scalar per alternative, and choices act on those scalars through α . Given the design and prior, the data therefore speak clearly about α — the within-menu η -contrasts identify it up to the scale fixed by the ν -endpoint convention — while saying little about how a given η splits into its (β, δ) parts. Strictly, the observed log-odds identify the products $\alpha(\eta_r - \eta_s)$; it is the endpoint convention that fixes the utility scale, together with the prior and design, that turns this into a sharp statement about α itself — as the posterior-contraction, parameter-recovery, and SBC diagnostics of §4 confirm.

Why α recovers cleanly anyway. Empirically, the two effects do not interfere: in parameter recovery (§4.3) α is recovered with low bias and well-calibrated intervals *regardless* of the wide, correlated spread in the (β, δ) posterior. Because α is identified from the within-menu choice-probability contrasts (Proposition 3.1) and not from η ’s internal (β, δ) decomposition, the indeterminacy in how η splits into beliefs and utilities leaves the α estimate essentially untouched. We report this as an observed feature of the recovered posterior rather than derive it from a separability argument.

This is a *structural* feature of decisions under uncertainty: beliefs and utilities enter only through expected utility, and uncertain-choice data alone supply little information for separating them.

3.6 From the weakly-informed (β, δ) to the extended model

That uncertain choices barely inform (β, δ) motivates the extended model of §5. Before getting there, §4 shows that the practical consequences — wide credible intervals on (β, δ) , tight intervals on α — are directly visible in a Stan implementation, and that marginal simulation-based calibration nonetheless passes for all three parameters (the *marginal-SBC demarcation*).

A Basic Computational Implementation: Model `m_0` in Stan

This section turns the abstract `m_0` model into an estimable Bayesian model and shows that the identifiability picture of §3 is directly visible in a Stan implementation: α recovers well, (β, δ) recover

poorly, and marginal simulation-based calibration (SBC) nonetheless passes for all three — the *marginal-SBC demarcation*.

4.1 Model specification

Data. M decision problems; R distinct alternatives with D -dimensional feature vectors w_r ; availability indicators $I_{m,r} \in \{0, 1\}$; observed choices y_m . A problem m presents $N_m = \sum_r I_{m,r}$ alternatives.

Parameters and priors.

$$\alpha \sim \text{Lognormal}(0, 1), \quad \beta_{k,d} \sim \mathcal{N}(0, 1), \quad \delta \sim \text{Dirichlet}(1, \dots, 1).$$

The $\text{Lognormal}(0, 1)$ prior on α (median 1, roughly 90% of mass in $(0.19, 5.18)$) is a *substantive* choice: it spans the full near-uniform-to-near-deterministic sensitivity range while concentrating mass near $\alpha = 1$. The prior shape co-determines which α values are well-measured in finite samples; we defend it via prior-predictive coverage in §4.2.

i Note for the reader (carried to §7). This foundational $\text{Lognormal}(0, 1)$ prior on α is *not* the application prior of §7, which recalibrates to $\text{Lognormal}(3.0, 0.75)$ for insurance and re-recalibrates for Ellsberg. The *likelihood* is unchanged across §4 and §7; only the *prior* is recalibrated.

Transformed parameters. $\psi_r = \text{softmax}(\beta w_r)$; the ordered utilities $v_k = \sum_{j < k} \delta_j$ via a cumulative sum (equivalently `upsilon = cumulative_sum(append_row(0, delta))`), so $v_1 = 0$ and $v_K = 1$; and $\eta_r = \psi_r^\top v$.

Likelihood. $y_m \sim \text{Categorical}(\text{softmax}(\alpha \eta_{[m]}))$, where $\eta_{[m]}$ collects the expected utilities of the alternatives available in problem m .

i The β gauge in the implementation. Softmax invariance to a common per-column row shift (§2.6) means β has only $(K - 1) \times D$ effective degrees of freedom. Rather than fix a reference row, `m_0` places an independent $\mathcal{N}(0, 1)$ prior on all $K \times D$ entries. The *likelihood* is flat along the gauge direction, but the Gaussian *prior* is not: its density decays as the common row shift moves β away from the origin, so the prior softly pins the gauge — effectively selecting the minimum-norm representative of each gauge orbit — and makes the posterior proper in that otherwise likelihood-invariant direction, without biasing the recovered ψ . Posterior summaries of β should therefore target row *contrasts* or the induced ψ , never absolute row levels.

4.2 Prior predictive analysis

The goal is to confirm that the priors permit a sensible *range* of choice behaviors before any data are seen. The summary statistic we monitor is the per-dataset **SEU-maximizer rate**: the fraction of choices that select the expected-utility-maximal available alternative under the simulated parameters. Drawing (α, β, δ) from the prior, simulating data via `m_0_sim.stan`, and tabulating this rate shows its prior-predictive distribution covering the full range from near-uniform choice (rate $\approx 1/\bar{N}$, the chance baseline) to near-deterministic SEU maximization (rate ≈ 1). The $\text{Lognormal}(0, 1)$ prior on α places substantial mass at both low (near-random) and high (near-deterministic) sensitivities, so neither regime is ruled out a priori.

4.3 Parameter recovery

Paradigm. Draw parameters from the prior via `m_0_sim.stan`, simulate data, fit `m_0.stan`, and compare posterior summaries to the known true values. We evaluate bias (≈ 0 ideal), credible-interval coverage ($\approx$ nominal 90%), and interval width / RMSE as precision measures.

α recovery. α recovers with low bias, well-calibrated 90% intervals, and useful precision; the headline display is a true-versus-estimated scatter with 90% credible intervals. This is the computational counterpart of Proposition 3.1: α is identified from η , and the data supply ample curvature to pin it.

β and δ recovery. By contrast (β, δ) show detectably wider intervals and intervals that narrow more slowly with sample size, together with correlated posterior-mean estimation errors — computed *across recovery iterations* between a representative weight entry $\beta_{1,1}$ and a representative increment δ_1 . We report this last item as an illustrative, non-gauge-invariant signature rather than a signed quantity: its sign is not robust (it depends on the true- α regime, since $\beta_{1,1}$ carries the row-shift gauge of §2.6), and it is the *presence* of across-replicate error coupling, not its direction, that matters here. This is the *computational manifestation* of the weakly-informed (β, δ) of §3.4: the multiplicative (β, δ) coupling realized as a correlated joint posterior and wide marginal intervals. Crucially, the poor (β, δ) recovery does *not* contaminate α — exactly as the empirical separation noted in §3.5.

4.4 Simulation-based calibration

Method. SBC draws a parameter from the prior, simulates data, fits the model, and computes the rank of the true value within the posterior draws; under correct calibration these ranks are uniform (Talts et al. 2018). We diagnose uniformity with rank histograms and the ECDF-difference plot with a simultaneous confidence band (Modrák et al. 2023), at $N_{\text{sbc}} = 999$, thinning 4, single chain, with the $L = N_{\text{sbc}}$ convention (Appendix D.3).

Result. At $N_{\text{sbc}} = 999$ the marginal rank distributions are consistent with uniformity for α , β , and δ . This run does **not** independently reproduce the weakly-informed (β, δ) finding of §§3–4.3, because that weakness is one of *joint informativeness and geometry* — a correlated, barely-contracted (β, δ) posterior that can itself be perfectly calibrated — to which marginal SBC is blind.

! **The marginal-SBC demarcation.** Two distinct points must be kept apart. *First*, marginal rank uniformity is necessary but not sufficient for *joint* posterior calibration: a joint miscalibration that left every margin uniform would not be caught here. *Second* — *and separately* — the correlated, barely-contracted joint (β, δ) posterior is a statement about weak *joint informativeness and geometry*, not about miscalibration: such a posterior can be exactly calibrated when the model and sampler are correct. What marginal SBC fails to reveal in our case is therefore the weak joint identification and contraction of (β, δ) , not a hidden joint calibration failure — we report no non-uniform joint SBC statistic. The positive contribution of SBC here is threefold: (i) **sampler/implementation validation** — uniform marginal ranks at $N_{\text{sbc}} = 999$ are strong evidence against a wide range of implementation and sampler defects (priors, simplex and ordered transforms, softmax curvature, HMC geometry) and establish *marginal* calibration; (ii) **marginal calibration** — per-parameter credible intervals are usable as marginal uncertainty summaries; and (iii) the **demarcation** itself — a precise statement of what marginal calibration can and cannot diagnose, which motivates a joint or projection-based rank statistic as future work (§8.3). We do *not* claim that marginal ranks certify correct sampling of the *joint* posterior, nor per-parameter calibration “fingerprints” or an `m_0`→`m_1` histogram improvement.

4.5 Section summary

`m_0` succeeds at recovering α — the parameter of primary interest — but the data leave (β, δ) only weakly informed, leaving the expected-utility *content* underdetermined at realistic n . This motivates the extended model of §5, whose implementation we take up in §6.

The Extended Abstract Model with Risky Choices

5.1 Enriching the choice domain — a two-step structural argument

The extended model (“`m_1`”) augments the uncertain-choice setup with risky alternatives whose probabilities are objectively given. The justification has two steps; Step 1 alone does *not* identify δ — Step 2 closes the argument.

i **Step 1 (β -free risky expected utility).** A risky alternative s carries an objective lottery $\pi_s \in \Delta^{K-1}$ over the K consequences, and its expected utility is $\eta_s^{(r)} = \pi_s^\top \mathbf{v}$, which does *not* pass through the belief-formation map. The risky block therefore supplies information about $\mathbf{v}$ (hence δ) that is *free of* β : the gradient of the risky log-likelihood with respect to δ does not also depend on β . By itself this supplies unconfounded gradients but not invertibility.

i **Step 2 (lottery diversity $\Rightarrow$ identification).** *Given sufficient lottery diversity* — the presented lotteries’ differences span the simplex tangent space, equivalently a spanning affinely independent subset of K lotteries with the menus linked into a connected contrast graph (§5.5), which requires $S \geq K$ — the β -free risky information *identifies* δ .

History (brief). Knight (1921) distinguished risk from uncertainty; von Neumann and Morgenstern (1944) axiomatized expected utility under risk; Savage (1954b) gave SEU under uncertainty.

Anscombe and Aumann (1963) combined the two by considering acts mapping states to lotteries, which lets one identify utilities from the risky margin and beliefs from the state margin. Our `m_1` uses a *related empirical strategy*: the connection is conceptual, not a direct application of the Anscombe–Aumann representation theorem. The important part is Steps 1+2, not the representation theorem.

5.2 The extended model

Augment the M uncertain problems with N risky problems built from S distinct lotteries π_s (objective simplexes over the K consequences). The model shares the same α and the same $\delta \rightarrow \mathbf{v}$ map across both blocks; risky expected utilities $\eta_s^{(r)} = \pi_s^\top \mathbf{v}$ depend only on δ , not on β . (The Stan code names π_s as `x_risky`; we gloss this inline in code excerpts and use π in the body.)

5.3 Preservation of the three sensitivity properties

The softmax properties of §2.3 are properties of the choice rule given *any* value function, so they apply unchanged to both uncertain and risky sub-problems. Therefore the interpretation of α as sensitivity-to-SEU-maximization is preserved in the extended model.

⚠ Single- α assumption. `m_1` asserts a *single* α governing both sub-problems — i.e., the agent’s responsiveness to expected-utility differences does not depend on whether probabilities are objective or subjective. This is a substantive, testable assumption, and it is essential to the §6.4 interpretation of the matched recovery contrasts: the quantity-versus-type reading of those contrasts requires a *common* α scale across blocks. If α differed across blocks, the matched comparison would conflate a choice-*type* effect with a *block-mix* effect on the effective α -precision. The generalization with separate $\alpha_{\text{unc}}, \alpha_{\text{risky}}$ (the companion model `m_2`) is named but not pursued here. **Consequently, every conclusion that compares α across the uncertain and risky blocks is reported as explicitly conditional on this single- α assumption**, supported where the data permit by a per-block SEU-maximizer-rate sanity check (§6.3); absent that check, the conditional phrasing is mandatory, not optional.

5.4 Identifiability of α in the extended model

i Proposition 5.1 (Identifiability of α). Proposition 3.1 holds unchanged when η is composed of both uncertain and risky alternatives: if some menu of positive design probability has non-constant η , then $\alpha > 0$ is identifiable from the choice-probability function.

5.5 Identifiability of δ in the extended model

i Proposition 5.2 (Identifiability of δ). If the presented lotteries’ differences $\{\pi_s - \pi_1\}$ span the $(K - 1)$ -dimensional simplex tangent space — equivalently, the design contains an affinely independent subset of K lotteries (requiring $S \geq K$ and a generic configuration) linked by a connected menu-contrast graph — then $\alpha > 0$ and δ are jointly and globally identifiable from risky-choice probabilities alone.

Intuition (full proof in Appendix B.3). A risky menu’s softmax depends on the linear functional $\pi_s^\top \mathbf{v}$. When the lottery differences span the simplex tangent space, the map $\mathbf{v} \mapsto ((\pi_s - \pi_1)^\top \mathbf{v})_s$ has rank $K - 1$, so the risky log-odds recover $\mathbf{v}$ — and jointly α — once the endpoint convention fixes the scale, hence δ . With $K = 3$ and $S = 15$ (Appendix D.0) the spanning condition is amply satisfied: 15 lotteries far exceed the $K = 3$ needed for a spanning affinely independent subset. The proof is a linear-algebra inversion.

5.6 β remains an irreducible nuisance

i Proposition 5.3 (β as an irreducible nuisance). With δ pinned by risky choices and α pinned by either block, the uncertain-choice probabilities identify only the expected utilities $\eta_r = \psi_r^\top \mathbf{v}$. For $K \geq 3$ these scalars do not determine the belief simplices ψ_r , so β is not point-identified beyond the level sets of $\beta \mapsto (\eta_r)_r$ together with the row-shift gauge of §2.6; it remains a nuisance even in the extended model.

Once δ (hence $\mathbf{v}$) is fixed, the uncertain-choice likelihood constrains β only through the scalar expected utilities $\eta_r = \psi_r^\top \mathbf{v}$. Because a scalar does not recover the $(K - 1)$ -dimensional simplex ψ_r when $K \geq 3$, the contribution of the risky block is to sharpen $\mathbf{v}$, not to recover β (Appendix B.4).

5.7 The conceptual upshot — and its finite-sample caveat

Enriching the choice domain with risky alternatives yields, *in principle*, identification of α (from either block) and of the utilities δ (from the β -free risky block, Step 1, plus lottery diversity, Step 2). It does *not* deliver the beliefs: β enters uncertain choices only through the scalar expected utilities and remains an irreducible nuisance for $K \geq 3$ (Proposition 5.3). And even the utility gain is an in-principle one — it implies estimability only *in the limit*. §6 asks whether the gain is realized at realistic finite sample sizes, and finds that for δ it largely is not — the cleanest illustration in the paper of the gap between identifiability and precise estimability.

A Basic Computational Implementation of the Extended Model: `m_1` in Stan

6.1 Model specification

`m_1` keeps the `m_0` parameters and priors — $\alpha \sim \text{Lognormal}(0, 1)$, $\beta_{k,d} \sim \mathcal{N}(0, 1)$, $\delta \sim \text{Dirichlet}(1, \dots, 1)$ — and adds N risky problems built from S distinct objective lotteries $\pi_s \in \Delta^{K-1}$ (Stan: `x_risky`). The log-likelihood is the sum of the `m_0` uncertain-choice term and a risky-choice term that uses $\eta_s^{(r)} = \pi_s^\top \mathbf{v}$ directly — in Stan, `eta_risky[i] = dot_product(x_risky[i], upsilon)`. The key structural contrast is that risky expected utilities depend only on $\mathbf{v}$ (hence δ), not on β (§5.1, Step 1).

6.2 Study design and the matched comparison

Design. We use the existing risky-choice configuration ($K = 3$, $S = 15$ lotteries; full constants in Appendix D.0), *not* a δ -optimal design. The matched design fixes one study design across conditions and slices the same simulated choices four ways:

Condition	Model	Uncertain M	Risky N	Total choices
A	m_0	25	—	25
B	m_0	50	—	50
C	m_1	25	25	50
D	m_1	50	50	100

The central test is **B vs C**: same total choice count, same true parameters per iteration, with only the model and the *type* of choice differing. The informative control is **A vs B**: doubling the *uncertain* block alone, holding the model fixed.

⚠ Statistical-power note. A recovery contrast at this scale needs enough iterations to pin effect *magnitudes*, not merely a *direction* of effect: with too few iterations the matched-count differences are too noisy to bound. We therefore report every §6.4 magnitude from a study of $n = 100$ iterations, each summarized with a bootstrap 90% CI over iterations.

What each reported magnitude is. Each of the $n = 100$ iterations runs the recovery loop of §4.3 on the matched design: draw one set of true parameters from the prior, simulate the four sliced choice sets (conditions A–D) under those parameters, fit the corresponding model in each condition, and record a per-iteration recovery metric — a posterior-mean RMSE against the known truth, or a posterior credible-interval width — for each condition. Because every condition within an iteration shares the *same* true parameters, the conditions are *paired*: a contrast such as B vs C or A vs B is formed within each iteration and then summarized across iterations by its paired-iteration median. The interval we report is a **bootstrap 90% CI over the $n = 100$ iterations** (10,000 resamples): we resample iterations with replacement, recompute the paired median on each resample, and take its central 90% range, so the CI measures how stable the aggregate contrast is across simulated datasets.

i Why a bootstrap rather than a posterior interval. The quantity summarized is a property of the *estimator and design under repeated simulation*, not a model parameter with a posterior — so a credible interval is not the right object. The Bayesian fit lives *inside* each iteration (every per-iteration RMSE or interval width comes from a full posterior); the bootstrap only quantifies the across-iteration variability of their paired median, distribution-free because those per-iteration paired differences are non-normal.

Sign convention. Throughout §6.4 a *positive* percentage denotes an *improvement* — a reduction in RMSE or in CI width — and its 90% CI is reported on the same scale, so a CI lying wholly above zero means a reliable improvement; a negative value denotes the effect running the other way (degraded precision).

What we do not claim. We do not engineer a design that makes the δ CI “materially” smaller than m_0’s; at $n = 100$ the matched-count δ CI-width gain is $\approx 0.8\%$ and δ RMSE is statistically unchanged. A δ -information-optimal lottery design — pitting, e.g., the certain intermediate consequence against a 50/50 mix of the extremes to triangulate each δ_k — is named as future work (§8.5), not a deliverable here.

6.3 Prior predictive analysis

On the combined (uncertain + risky) prior predictive, the SEU-maximizer rate again covers the full sensitivity range. Because `m_1` assumes a *shared* α across blocks (§5.3), we also report the prior-predictive SEU-max rate **per block** (uncertain vs risky) as a sanity check that the prior alone places plausible mass on both blocks under a common α . This is a statement about the *prior*, not a test of the shared- α assumption against data: confirming that risky and uncertain choices are mutually compatible with a single α would require a *posterior*-predictive block check, which we flag as the appropriate diagnostic for the single- α caveat (§6.4.1) rather than claim to have performed here.

6.4 Parameter recovery — what the matched comparison actually shows

6.4.1 α : no per-choice advantage from the risky block

At matched total choice count (B vs C), swapping in the β -free risky block yields **no** α -RMSE improvement: -8.1% , bootstrap 90% CI $[-22.0, +6.8]$ — the point estimate is slightly *worse* (it favors `m_0`) and the interval straddles zero. The CI-width measure agrees (paired median -1.4% , improved in only 44% of iterations; Wilcoxon signed-rank $p \approx 0.39$). Here and below the Wilcoxon signed-rank test is a paired, distribution-free test of whether the per-iteration paired differences between conditions are centered at zero; we report it alongside the bootstrap CI because it makes no normality assumption about those differences and a small p would indicate a systematic shift in one direction across iterations.

The informative control is A→B: doubling the *uncertain* block alone cuts α RMSE by **27.1%** (90% CI $[+15.0, +38.5]$; CI-width paired median $+14.0\%$, improved in 78% of iterations). Read together, the two contrasts show that α precision at realistic n is driven by data *quantity*, not by the *type* of choice: per choice, uncertain choices are at least as informative for α as risky ones, so the β -free risky block confers no special per-choice advantage. We draw out the methodological reading in §6.4.4.

i Conditional on the single- α assumption. Comparing α precision across the uncertain and risky blocks presupposes a common α scale across blocks (§5.3). The per-block *prior*-predictive SEU-maximizer-rate check of §6.3 is the available prior-level sanity check — a genuine data-level test of shared α would be posterior-predictive — so the quantity-versus-type reading is reported as explicitly conditional on this assumption, not asserted outright. The block-specific generalization (`m_2`) is out of scope.

6.4.2 δ : the payoff is essentially nil

At matched choice count (B vs C), `m_1` narrows the δ CI width by only $\approx 0.8\%$ (paired-iteration median; bootstrap 90% CI $[0.6, 1.2]$; narrower in 72% of iterations; Wilcoxon signed-rank $p \approx 1.2 \times 10^{-6}$) and leaves δ RMSE statistically unchanged (0.3% , 90% CI $[-4.0, +4.5]$). The CI-width effect is *real but practically negligible*. The reading: δ is identifiable in `m_1` (§5.5, hence estimable *in the limit*) but **not precisely estimable at realistic sample sizes** in this design.

6.4.3 β

β recovery in $\mathbf{m}_1$ is improved relative to $\mathbf{m}_0$ but remains subject to the additive row-shift gauge (§2.6; Appendix B.4); summaries target contrasts and induced ψ , not absolute row levels.

6.4.4 The methodological lesson — two phenomena

The matched comparison yields two separable substantive lessons, which we keep apart.

(i) **For δ — the canonical lesson, sharpened.** The principle is that *identifiability (hence estimability in the limit) does not entail precise estimability at realistic sample sizes*. δ is identifiable in $\mathbf{m}_1$ (§5.5), but the information about δ contributed by 25 risky choices at moderate α is small: the matched B→C comparison shrinks the δ CI width by $\approx 0.8\%$ and moves δ RMSE not at all. In-principle identification buys *essentially nothing* at the design sample size — the identifiability-versus-estimability gap in its starkest form.

(ii) **For α — finite- n precision tracks quantity, not type.** One might expect the $\mathbf{m}_1$ α gain to be a per-choice advantage of the β -free risky block; at $n = 100$ there is no such advantage (B→C null). What sharpens α is simply *more data of the same kind* (A→B: a 27.1% RMSE reduction). The correct lesson is *not* “a special structure sharpens an already-identified parameter”; it is that **in-principle identifiability tells you nothing about where finite- n precision comes from** — here it comes from quantity, not type. Phenomena (i) and (ii) are distinct.

Table 3: The matched-design contrast. All percentages are paired-iteration medians / bootstrap-over-iteration 90% CIs from the $n = 100$ study (10,000 resamples).

	δ in $\mathbf{m}_1$	α in $\mathbf{m}_1$
Identifiable in $\mathbf{m}_0$?	yes (jointly weak)	yes
What §5 promises	becomes identifiable via the β -free risky block	already identified
Matched-count payoff (B→C)	CI-width +0.8% [0.6, 1.2]; RMSE null	none: −8.1% [−22.0, +6.8]
Where finite- n precision comes from	not realized at design n	data <i>quantity</i> (A→B: +27.1% RMSE reduction [+15.0, +38.5]), not choice <i>type</i>
Phenomenon	identifiability $\neq$ precise estimability at realistic n	in-principle identification $\neq$ finite- n gain; precision tracks quantity, not type

6.5 SBC for $\mathbf{m}_1$

Rank histograms and ECDF-difference plots for α , β , δ are consistent with uniformity. We do **not** claim an $\mathbf{m}_0 \rightarrow \mathbf{m}_1$ δ -calibration improvement (e.g., a flatter histogram): marginal SBC is uniform for δ in *both* models, because the relevant (β, δ) weakness is in the *joint* posterior (the **marginal-SBC demarcation** again, §4.4). The appropriate diagnostic is a joint or projection-based rank statistic (§8.3). The usual single-chain SBC caveats apply: thinning, single-chain Monte-Carlo error, and sample-size driven rank-resolution all bound what the histograms can detect.

Illustrative Application: SEU Sensitivity in LLM Decisions

7.1 Why include an application here

The methodology of §§2–6 is only useful insofar as it does real evaluative work on real choice data. This section runs the full workflow end-to-end on a 2×2 factorial design — two LLMs crossed with two task families — to demonstrate three things: (i) the workflow scales from the $N \approx 100$ simulated choices per condition of §§4/6 to $N \approx 300$ real-LLM choices per condition without redesign; (ii) prior recalibration is a routine, principled per-study step rather than a redesign of the model; and (iii) the framework registers a structured comparative effect when the posterior supports one and withholds support when it does not — where “withholds support” means *inconclusive at the achievable resolution*, not an established zero (§7.5.2). Point (iii) is the methodological-value claim: an evaluation framework that *always* reports an effect would be a scoreboard, not an instrument.

! Reading guide (construct-validity layering; read first). Three distinct claims must not be conflated, and this layering frames everything below:

1. **Within-model consistency** — the model fits the data adequately (validated in §7.4).
2. **Comparative claims under shared design** — α differs systematically across conditions that share prior, features, and choice model (the §7.5 reading).
3. **Absolute claims about EU rationality** — an LLM’s α value certifies a context-free rationality level.

Only layers 1 and 2 are licensed by this paper. §7.6.3 deepens this; it is stated here so the reader reaches the §7.5 headline already holding it.

i Reporting template (applied uniformly to all four cells). Every LLM $\times$ task cell is reported with the *same* quantities, to pre-empt cherry-picking: per-condition α posterior medians with 90% CIs; the global slope $\Delta\alpha/\Delta T$ median, 90% CI, and $P(\text{slope} < 0)$; $P(\text{strict monotone decrease across all levels})$; and the cross-LLM comparison probability. Cross-task differences are read descriptively (the 2×2 pattern of §7.5.5) rather than as a single formal contrast probability, because the two tasks differ in K and in their separately calibrated α priors and so do not share one design. No cell omits a quantity.

7.2 The 2×2 design

LLMs. GPT-4o (OpenAI) and Claude 3.5 Sonnet (Anthropic).

Task families. (i) *Insurance claims triage* over a 30-claim pool, with $K = 3$ consequences — the three investigator-agreement outcomes (neither, one, or both investigators agree the claim warrants investigation); and (ii) *Ellsberg-style urn gambles* over a 30-gamble pool, with $K = 4$ consequences (\$0, \$1, \$2, \$3 payouts), organized into three ambiguity tiers (Tier 1 unambiguous, Tier 2 moderately ambiguous, Tier 3 high-ambiguity in the spirit of Ellsberg (1961)). In both tasks the *alternatives* presented in a problem are the pool items themselves (claims or gambles), and the menu size $N_m \in \{2, 3, 4\}$ is drawn at random per problem.

Per cell. Five sampling-temperature conditions; $M \approx 100$ base problems $\times$ 3 position-counterbalanced presentations = 300 choices per condition (constants in Appendix D.5). Temperature grids differ by provider: GPT-4o {0.0, 0.3, 0.7, 1.0, 1.5}; Claude {0.0, 0.2, 0.5, 0.8, 1.0}

(Anthropic-API range constraint). The unequal grids complicate direct slope-magnitude comparison across LLMs, though not the qualitative reading; we flag this again at §7.5.2a.

Feature pipeline. Both task families use the *same* two-stage construction. *Insurance*: a per-claim LLM assessment, then choice over those assessments; assessment text embedded via `text-embedding-3-small`; pooled-PCA projection to $D = 32$. *Ellsberg*: the identical pipeline applied to gambles — a per-gamble LLM assessment of each urn (a short free-text analysis of the payoff probabilities and ambiguity), then choice over those assessments, embedded and pooled-PCA-projected to $D = 32$. The objective urn composition is *not* entered into the model directly: in both tasks the feature vector w_r is an embedding of an LLM-generated assessment, and the choice is made over those assessments. Both applications therefore measure the *whole assessment-and-choice pipeline*, not the choice stage in isolation; the two cells differ in the decision domain (claims vs. gambles) and in K (3 vs. 4), not in which stages the temperature lever touches (we return to this at §7.6.5). Position counterbalancing addresses a position-bias problem identified in preliminary elicitation runs; unparseable responses are recorded as missing rather than coerced to a default, and missing rates are negligible across all 20 conditions.

7.3 The model fit (m_01 / m_02)

The fitted model is structurally identical to m_0 of §4 but with a Lognormal(μ, σ) prior on α calibrated to each application’s design and consequence space: m_01 for the insurance task ($K = 3$) and m_02 for the Ellsberg task ($K = 4$). The two differ only in that α prior.

Prior calibration anchor. The calibration target is the prior predictive SEU-maximizer selection rate (fraction of simulated problems on which the simulated agent picks the EU-maximal available alternative). The foundational Lognormal(0, 1) prior of §4.1 places excessive mass on near-random regimes and, at the application’s R and K , on tails that cause softmax overflow under the study design. A grid search over twelve Lognormal hyperparameter pairs selects Lognormal(3.0, 0.75) for the insurance task ($K = 3$): median $\alpha \approx 20$, 90% interval $\approx [5.5, 67]$, implied SEU-max rate $\approx 78\%$. For the Ellsberg task ($K = 4$) the prior is *re-calibrated* by the same procedure, selecting Lognormal(3.5, 0.75) (median $\alpha \approx 33$, 90% interval $\approx [10, 124]$, implied SEU-max rate $\approx 76\%$). The recalibration is needed not because the random-choice baseline changes — that baseline is set by the menu size, not the consequence count K , and the menu size $N_m \in \{2, 3, 4\}$ is identical across the two tasks, so a uniform-random chooser selects the SEU-maximizer with the same $1/N_m$ probability either way — but because the larger consequence space ($K = 4$) changes the spread of expected utilities across alternatives, and hence the softmax curvature, so a higher α is required to reach the same prior-implied SEU-max rate.

i Only the prior is recalibrated, not the likelihood. The α -identifiability proposition of §3.3/§5.4 and the weakly-informed (β, δ) finding of §3.4 carry over unchanged from m_0 to m_01. The application inherits the identifiability story of §§3 and 5 without re-derivation. This foundational-vs-application prior distinction is the one flagged in §4.1: same likelihood, different α prior.

7.4 Validation at the application’s scale

The workflow of §§4.2–4.4 is re-run at the application’s scale ($M = 300$, $K = 3$ or 4 , $D = 32$, $R = 30$, 20 recovery iterations per cell).

Parameter recovery for α . Anchored on relative metrics, because true α lies on a wide multiplicative scale under the calibrated prior: relative bias within $\pm 10\%$ of the mean true value, relative RMSE well below 25%, and 90% CI coverage at nominal. As throughout, (β, δ) recovery remains weak — the §3 picture is unchanged — but the cross-condition comparison is a claim about α , and α is fit for purpose.

SBC for α . SBC validates the *(prior, likelihood, sampler)* triple and is conditionally independent of which agent generated the held-out empirical data.

i Remark (SBC reuse). This is a workflow convention rather than a proved proposition: if two fitted studies use the same Stan model, the same prior hyperparameters, and the same feature pipeline, an SBC result established for one can be reused for the other without re-running — which is what spares us a second, expensive calibration for the matched cell. *Application:* SBC for α under Lognormal(3.0, 0.75) and the insurance likelihood is established once in the GPT-4o $\times$ insurance report and inherited by the Claude $\times$ insurance study. The Ellsberg studies differ in K and in the α prior and so receive **fresh** SBC under the $K = 4$ recalibrated prior.

Qualification. SBC certifies the *(prior, likelihood, sampler)* triple at the model’s dimensions (K, D, R) ; it fixes a design when simulating, so strictly it speaks to the particular realized feature matrix used. Because GPT-4o and Claude generate different assessments — and hence different w — the insurance inheritance is an approximation across two provider-specific designs of the same dimensions, not an identity; the per-cell parameter-recovery diagnostics above are what confirm that calibration holds at each realized design.

MCMC diagnostics. Across all 20 factorial fits, α — the parameter the §7.5 temperature analysis rests on — is **never** $\hat{R}$ -flagged, and ESS is satisfactory in every fit, so the per-condition α posteriors are a sound basis for the cross-condition comparison. Divergent transitions are negligible ($\leq 0.15\%$ per fit, $\leq 6/4000$; 34 total), though spread across 12/20 fits in both tasks and both providers rather than confined to the highest GPT-4o temperatures. $\hat{R} > 1.01$ appears in 4/20 fits as marginal exceedances on a minority of high-dimensional components, **confined to the weakly-informed nuisance parameters** — β , δ , and the global length- K utility vector v (pinned only up to the §3 gauge) — together with the per-trial latents η and ψ in the harder $K = 4$ GPT-4o $\times$ Ellsberg fits (plus one Claude $\times$ insurance fit, η only). This *corroborates* the §3.4/§6.4 weakly-informed (β, δ) finding — the poorly-mixing parameters are exactly the ones recovery flags as weakly informed — and is not a defect in α inference.

Posterior predictive checks. Three complementary summaries (log-likelihood, modal-choice frequency, mean predicted probability of the chosen alternative) yield posterior predictive p -values in $[0.32, 0.66]$ across all 20 fits (60/60 in band, mean ≈ 0.46) — no evidence of systematic misfit. PPC adequacy is a *necessary* (not sufficient) condition for the per-condition α posteriors to be a credible basis for the cross-condition comparison; absent it, the comparison should be set aside.

7.5 Results — the 2×2 cross-LLM $\times$ cross-task picture

7.5.1 GPT-4o $\times$ insurance

Posterior medians of α are highest at $T = 0.0$ and lowest at $T = 1.5$, intermediate temperatures monotone-ish in between. The global slope $\Delta\alpha/\Delta T$ has median ≈ -31 , 90% CI $\approx [-66, -8]$, $P(\text{slope} < 0) \approx 0.99$. The probability of strict monotonic decrease across all five levels is only ≈ 0.12

(driven by overlap between $T = 0.3$ and $T = 0.7$); collapsing those two raises it to ≈ 0.38 . The headline *directional* claim is well-supported; fine-grained adjacent-step orderings are not.

7.5.2 Claude $\times$ insurance

The framework **does not detect** a monotonic α -temperature pattern (note the careful phrasing: *does not detect*, not *establishes the absence of*). Posterior medians are $\approx \{71, 53, 73, 71, 55\}$ at $\{0.0, 0.2, 0.5, 0.8, 1.0\}$. Global slope: median ≈ -3.6 , 90% CI $\approx [-54, 39]$, $P(\text{slope} < 0) \approx 0.56$; $P(\text{strict monotone decrease}) < 0.01$. The oscillation in the medians is consistent with posterior noise around a roughly flat function, not a substantive non-monotonic response.

⚠ Minimum-detectable-effect (calibrating the Claude null). The design could resolve a temperature-on- α slope at $P(\text{slope} < 0) \geq 0.95$ only if its magnitude were $|\Delta\alpha/\Delta T| \gtrsim 36$ α -units per unit temperature (three independent estimators agree: analytic-Gaussian 36.3, empirical-quantile 34.7, constant-CV Monte-Carlo 36.0; ≈ 41 – 43 at the stricter $P \geq 0.975$). That floor is ≈ 10 – $12\times$ the observed $|\text{slope}|$ — whether taken as the draw-level posterior median that the power calculation operates on (≈ 2.9) or the report-level headline summary (≈ 3.6); the two differ only by which posterior draw set is summarized, and the null sits an order of magnitude below the floor either way — and ≈ 0.53 of the grand-mean α (≈ 67.5) — an end-to-end α change of ≈ 36 over $T \in [0, 1]$, roughly halving α . So the Claude null is *no effect at the achievable resolution*, **not** a measured zero; even a GPT-4o-sized slope (≈ 31 , full-grid reference) sits below this floor ($0.86\times$). See the power curve (Fig 7.5.2). This calibration prevents the §7.6.1 “declines to support” framing from being read as a positive no-effect claim.

7.5.2a Cross-LLM comparison — pattern first, number second

A formal cross-study comparison gives $P(\text{GPT-4o slope} < \text{Claude slope}) = 0.817$ on the full grids; restricting GPT-4o to Claude’s $T \leq 1.0$ ceiling (re-summarizing with $T = 1.5$ dropped) gives 0.824 — the directional LLM contrast is **robust to the unequal-grid confound** (GPT-4o spans $T \in [0.0, 1.5]$; Claude spans $[0.0, 1.0]$). We report the Claude-grid-restricted number alongside the full-grid one and make the *qualitative pattern* (monotone-decline vs. no-monotone) the headline, not the numeric inequality. Both between-LLM probabilities (~ 0.82) are weaker than GPT-4o’s own within-cell $P(\text{slope} < 0) > 0.98$ because the cells are fitted independently. “Temperature” is not the same instrument across providers (§7.6.2(a)).

7.5.3 GPT-4o $\times$ Ellsberg

The monotonic decline of α with temperature is reproduced on the Ellsberg task. α medians $\{110.4, 106.9, 99.5, 84.0, 52.2\}$ at $T = \{0.0, 0.3, 0.7, 1.0, 1.5\}$ (90% CIs $[74.4, 167.2]$ / $[72.6, 163.8]$ / $[65.4, 154.6]$ / $[57.1, 126.1]$ / $[35.5, 80.3]$); global slope $\Delta\alpha/\Delta T$ median -38.4 , 90% CI $[-72.1, -10.0]$, $P(\text{slope} < 0) 0.984$; $P(\text{strict monotone } \downarrow) 0.090$ (a noisy decline, not step-wise).

7.5.4 Claude $\times$ Ellsberg

No monotonic α -temperature pattern. α medians $\{85.4, 56.1, 82.3, 53.0, 66.4\}$ at $T = \{0.0, 0.2, 0.5, 0.8, 1.0\}$ (90% CIs $[58.9, 127.2]$ / $[38.2, 84.1]$ / $[52.3, 135.3]$ / $[36.0, 80.5]$ / $[45.3, 99.8]$) — the medians oscillate rather than decline; global slope $\Delta\alpha/\Delta T$ median -18.8 , 90% CI $[-65.3, 24.5]$,

$P(\text{slope} < 0) 0.766$; $P(\text{strict monotone } \downarrow) 0.0085$. As with Claude $\times$ insurance, this is *does not detect*, not *establishes absence* (cf. the §7.5.2 MDE discipline).

7.5.5 The 2×2 reading

Table 4: Per-cell global-slope $\Delta\alpha/\Delta T$ posteriors (median, 90% CI, $P(\text{slope} < 0)$). GPT-4o shows the temperature- α decline in both task families; Claude shows it in neither.

	Insurance	Ellsberg
GPT-4o	slope -31 $[-66, -8]$, $P_{<0} 0.99$	slope -38.4 $[-72.1, -10.0]$, $P_{<0} 0.98$
Claude 3.5	slope -3.6 $[-54, 39]$, $P_{<0} 0.56$	slope -18.8 $[-65.3, 24.5]$, $P_{<0} 0.77$

In this 2×2 , the LLM $\times$ task factorial isolates a clean **LLM effect**: GPT-4o exhibits the temperature- α relationship across both task families, Claude exhibits it in neither. *In this 2×2* , task family does not appear to be the relevant axis; the LLM does. With only two task families and two LLMs this is a **pattern statement**, not a **factorial generalization** (§7.6.2(d)). It is best presented as a 2×2 forest plot of the per-cell global-slope posteriors.

7.6 What the application demonstrates — and what it does not

7.6.1 What it demonstrates

- (i) The full workflow of §§4 and 6 runs end-to-end on real LLM choice data, at the application’s scale, with the α inference and PPC summaries clean and the only flagged diagnostics confined to the weakly-informed nuisance parameters (§7.4). (ii) Prior recalibration via the prior predictive SEU-maximizer rate is a routine, principled per-study step. (iii) The framework detects a structured comparative effect in two of four LLM $\times$ task cells (GPT-4o, both tasks) and **does not detect** one in the other two (Claude, both tasks). This last point is the methodological-value claim — but stated carefully: the Claude cells are *inconclusive nulls* (§7.5.2), so the licensed claim is “an instrument that does not manufacture an effect where its own diagnostics cannot resolve one,” **not** “an instrument that certifies the absence of an effect.” The minimum-detectable-effect statement of §7.5.2 is what keeps this claim resting on calibrated resolution rather than an unsupported null.

7.6.2 What it does not demonstrate

(a) **That sampling temperature is a portable behavioral instrument.** The Claude cells do not reproduce the GPT-4o pattern under the same task and choice model; but the minimum-detectable-effect analysis of §7.5.2 shows that even a GPT-4o-sized slope would sit below this design’s resolution floor for Claude, so the non-reproduction is *inconclusive*: it does not establish that temperature behaves differently across providers, only that any Claude-side effect, if present, is below what these data can resolve. (b) **That GPT-4o is “more rational” than Claude in any context-free sense.** The licensed reading of α here is a *within-design comparative* one, not an absolute-rationality ranking. (c) **Anything about ambiguity aversion in the Ellsberg study.** The SEU+softmax model is, by assumption, of the wrong form to distinguish ambiguity-driven

choice from EU-driven choice; testing for ambiguity aversion requires a model with explicit ambiguity-attitude parameters (e.g., Gilboa and Schmeidler (1989); α -MEU à la Ghirardato, Maccheroni, and Marinacci (2004)), and pooling across the three ambiguity tiers as the application does would yield a mixture rather than identify a mechanism. Tier-stratified α is named as future work, not a paper claim. **(d) That LLM identity is the only relevant axis in general.** With only two task families and two LLMs, the 2×2 supports a pattern statement, not a factorial generalization.

7.6.3 Construct-validity layering (reaffirmed)

The three-layer reading guide of §7.1 — (1) within-model consistency, (2) comparative claims under shared design, (3) absolute claims about EU rationality, with only (1) and (2) licensed — is the single most important reading instruction for this section. The §7.5 cross-condition contrast is a layer-(2) claim; it is *not* a layer-(3) certification of either LLM’s context-free rationality.

7.6.4 Ellsberg: do not adjudicate the normative question

The Ellsberg stimuli carry historical weight that does not reduce to the behavioral “ambiguity aversion” reading: alternative readings include Ellsberg’s own normative defense (Ellsberg 1961, 2001) and Levi’s reading as a violation of the completeness axiom rather than the sure-thing principle (Levi 1980, 1986). A study that takes SEU as its *measurement device* cannot use the resulting estimates to vindicate or refute SEU as a *normative* standard. (See Camerer and Weber (1992) and Trautmann and Kuilen (2015) for the broader behavioral-ambiguity literature.) The paper’s contribution lies elsewhere.

7.6.5 Both applications measure the whole pipeline

Neither application isolates a single cognitive stage, and it is worth being precise about what α actually measures. Each trial — insurance or Ellsberg — is produced by a *composed system*: a first LLM call assesses the stimulus (a free-text claim, or an urn gamble) in natural language, an embedding model maps the resulting text to a vector, a PCA step reduces it to the feature space the choice model reads, and a second LLM call then chooses among the assessed alternatives (§7.2). The fitted sensitivity α is a property of that whole assembly — the assessment and choice calls together with the deterministic feature pipeline — and not of the choice stage taken on its own. The estimand is therefore the SEU sensitivity *of the pipeline as configured*, and the temperature lever acts on every LLM-mediated component of it at once. This is the appropriate object of study for the methodological claim — the workflow recovers a well-calibrated α for a realistic end-to-end system — but it means that a within-cell change in α cannot, by itself, be attributed to any one stage. The GPT-4o monotonicity is thus consistent with temperature affecting (i) the assessment stage, (ii) the choice stage, or (iii) both.

The Ellsberg cell does *not* remove the assessment layer. It applies the *same* assess–embed–choose construction to a different decision domain ($K = 4$ monetary gambles in place of $K = 3$ claim-triage outcomes); the objective urn composition enters only through the LLM’s free-text assessment, never as a structural input to the model. It is therefore a *cross-domain replication* rather than a stage-isolation design. What it establishes is correspondingly more modest, but still useful: that the GPT-4o temperature pattern is not an artifact of the particular insurance content or its assessment prompts, since a comparable effect reappears when the same pipeline is pointed at a structurally different task. It does *not* decompose the effect by stage — because the domain and the assessment prompt change together with everything else, the Ellsberg cell cannot say whether the insurance

effect originates in the assessment stage, the choice stage, or both. The two cells together show that the effect travels across domains; isolating its locus within the pipeline would require dedicated stage-ablation designs (assessment-only versus choice-only manipulations) that lie beyond this paper’s scope.

7.7 What the application motivates for follow-up work

Design-induced cross-condition correlation. The five temperature conditions in each cell draw from a single *fixed set* of $R = 30$ alternatives — the same claims (or gambles) at every temperature, though their LLM assessments and embeddings are regenerated independently at each temperature (§7.2). Whatever is idiosyncratic about that particular set — its embedding geometry, EU spread, or typical best-vs-second-best gap — is a property of *this* pool rather than of the population of pools the design might have drawn, so the cell’s contrasts speak to the realized pool and need not transport to another. Two concerns follow that independent per-condition $\mathbf{m_01}$ fits do not address: the cross-condition contrasts may not generalize beyond the realized pool, and the pool may interact with temperature — a pool-by-temperature effect that a single-pool design cannot separate from the main effect. The principled fix is the hierarchical extension $\mathbf{h_m01}$: $\log \alpha_c = \gamma_0 + \gamma_1 T_c + \eta_c$ with a small cell-level random effect, which estimates the temperature–sensitivity relationship as a single regression slope with one calibrated uncertainty statement and partially pools the five conditions; its multi-pool generalization (the companion paper) additionally lets the item pool be modeled as a source of variation rather than treated as a fixed idiosyncrasy.

Companion paper. A planned alignment-study companion paper (multi-LLM $\times$ multi-prompt factorial) will introduce $\mathbf{h_m01}$ as the analysis vehicle and apply it to the alignment-study data. That paper is out of scope here; the present paper references it once, as the natural next step for which $\mathbf{h_m01}$ was developed.

Tier-stratified α for the Ellsberg studies, and a non-SEU comparator (e.g., a small α -MEU instrument) that could be paired with the SEU instrument to triangulate ambiguity-driven vs. EU-driven choice, are flagged as future work in the application reports and remain out of scope here.

Discussion

8.1 What “sensitivity to SEU maximization” means, restated

Sensitivity to SEU maximization is the *disposition* of an agent committed to SEU maximization to act in accordance with that commitment. The model decomposes choice behavior into three conceptually distinct objects: (i) the agent’s beliefs (β , hence the subjective probabilities ψ); (ii) the agent’s utilities (δ , hence the ordered utilities $\mathbf{v}$); and (iii) the agent’s sensitivity α to the expected-utility ranking those beliefs and utilities induce. Beliefs and utilities are *what* the agent is committed to; α is *how reliably* the agent acts on those commitments. The three softmax-choice properties of §2.3 — monotonicity in α , the optimization limit, and the uniform-choice limit — are exactly what license reading α as a graded measure that runs from near-uniform choice ($\alpha \rightarrow 0$) to deterministic SEU maximization ($\alpha \rightarrow \infty$).

This reading returns to Isaac Levi’s distinction between an agent’s *commitment* to a standard and the agent’s *performance* relative to it (Levi 1980, chap. 1), introduced in §2.5. One can be committed to a standard one fails to live up to: most of us are committed to the laws of arithmetic yet occasionally make calculation errors, and those errors are lapses of performance rather than

rejections of the standard. Taking SEU theory to specify the agent’s normative commitment, α is exactly the agent’s *tendency to perform in accordance with that commitment*.

Two clarifications keep this borrowing honest. First, Levi was no defender of SEU as the standard of rational choice; on the contrary, he developed an influential generalization of it that admits indeterminate (imprecise) probabilities and utilities (Levi 1986), and his substantive decision theory therefore falls outside the real-valued template of this paper (§1.4). What we borrow is his *meta-level* distinction between commitment and performance, not his account of the standard itself. Second, that distinction is of interest even once SEU is granted as the relevant standard. Consider the SEU violations documented by Kahneman and Tversky (Kahneman and Tversky 1979; Tversky and Kahneman 1992): do they reveal failures of *performance* by agents committed to SEU, or an *absence of commitment* to SEU as a standard? Our framework does not settle that question, but it does make it tractable: *under the assumption* that the agent is committed to SEU, α is an estimate of the strength of that commitment — the agent’s disposition to fulfil it.

8.2 The methodological role of identifiability analysis

Identifiability is a property of the *likelihood*: it tells us which parameters could in principle be recovered from choice data. It does *not* tell us how much data, or how good a design, is needed to recover them precisely. The governing principle is:

! *Identifiable (hence estimable in the limit) $\neq$ precisely estimable at realistic sample sizes.*

The m_0 / m_1 contrast illustrates *both* sides of this distinction, and the paper is careful to keep the two sides apart (§6.4.4):

- **For δ — identifiability without precise estimability.** In m_1 , δ is globally identifiable from the β -free risky block under lottery diversity (§5.5). Yet at the design’s sample sizes the matched-count payoff is essentially nil: a δ CI-width reduction of $\approx 0.8\%$ (paired-iteration median; bootstrap 90% CI [0.6, 1.2]; Wilcoxon $p \approx 1.2 \times 10^{-6}$) and no change in δ RMSE. The effect is statistically real but practically negligible — the identifiability-versus-estimability gap in its starkest form.
- **For α — in-principle identifiability is silent about where finite- n precision comes from.** α is identifiable from η in both models (§3.3, §5.4). The matched recovery study shows that at realistic n its precision is governed by data *quantity*, not by the *type* of choice: doubling the uncertain block alone sharpens α ($A \rightarrow B$: RMSE reduced 27.1%, 90% CI [+15.0, +38.5]), whereas swapping in an equal count of β -free risky choices does not ($B \rightarrow C$: -8.1% , 90% CI [−22.0, +6.8], straddling zero). An in-principle identification argument predicts neither of these finite- n facts.

8.3 The methodological role of computational validation

Three checks should be standard practice for any choice-theoretic measurement instrument, and the paper treats them as such. *Prior predictive checks* (§4.2, §6.3) expose what a prior implies before any data are seen; in this setting the SEU-maximizer-rate statistic makes the α prior’s behavioral content legible and turns per-study prior recalibration into a principled, repeatable step (§7.3). *Parameter recovery* and *simulation-based calibration* (SBC) together test whether the posterior recovers known truths — recovery for bias, precision, and interval coverage; SBC for calibration of the whole (*prior, likelihood, sampler*) triple.

A specific caution emerges from our results, which we name so it is citable rather than discursive:

⚠ The marginal-SBC demarcation. Two points must be kept apart. *First*, marginal rank uniformity is *necessary but not sufficient* for *joint* posterior calibration. *Second* — and *distinctly* — the correlated, barely-contracted joint (β, δ) posterior reflects *weak joint informativeness* (a broad, trade-off-shaped likelihood geometry), **not** a calibration failure: a correctly specified model and sampler can be exactly calibrated and still return such a posterior. Marginal SBC passes for α , β , and δ in both `m_0` and `m_1` (§4.4, §6.5); what it does *not* do is reveal the weak joint identification of (β, δ) at realistic n (§3.4). We report no non-uniform joint SBC statistic and claim none: the demarcation is about what marginal ranks can and cannot *diagnose*, not evidence of a hidden joint miscalibration.

The positive contribution of SBC here is therefore threefold: (i) sampler/implementation validation — uniform marginal ranks at $N_{\text{sbc}} = 999$ are strong evidence against a broad class of implementation and sampler defects (simplex parameters, ordered-utility transform, softmax curvature) and support marginal calibration, though they do not by themselves certify that the encoded *joint* posterior — including the (β, δ) dependence — is sampled correctly; (ii) per-parameter marginal calibration, so the credible intervals are usable as marginal uncertainty summaries; and (iii) the demarcation itself, which clarifies *which* identifiability properties marginal SBC can and cannot diagnose. When parameters trade off in a correlated joint posterior, a *joint* or *projection-based* rank statistic is the appropriate diagnostic — a concrete next step for the SBC literature that this paper’s null between `m_0` and `m_1` motivates.

8.4 Why α is the primary quantity of interest

A central methodological claim of the paper is that α can be measured precisely *even when* (β, δ) cannot be — a separation the recovery study and application deliver *empirically*, under the paper’s priors and designs, rather than one guaranteed by the conditional identification proposition alone. The mechanism is that beliefs and utilities enter the choice probabilities only through the expected-utility vector η , and the log-odds scale with differences in η at rate α (§3.5). Proposition 3.1 identifies α only *conditional* on η ; in the uncertain-choice model η is itself generated by the unknown (β, δ) , and near the linear regime of the belief softmax a change in the spread of η can trade off with α in the observed log-odds. What makes α sharply estimable in practice is therefore an empirical fact about the calibrated prior and the realized designs — the posterior concentrates the α -bearing log-odds scale while leaving the (β, δ) trade-off broad (§3.4) — not a structural guarantee read off the conditional proposition. The §7 application instantiates the claim: the cross-condition comparison is a claim about α , α is sharply and calibratedly recovered in `m_0` / `m_01`, and the weak identifiability of (β, δ) does not undermine that comparison.

i Interpretive caveat (model-conditional reading of α). A low recovered α is *not*, in a model-free sense, “low SEU sensitivity.” The measured α is a parameter of a specific likelihood — a softmax over SEU values under the parameterization of §3. A low α is observationally equivalent to several non-SEU alternatives (the agent maximizes a different value functional, uses a non-softmax stochastic rule, has heterogeneous within-session preferences, and so on). The paper’s claim is therefore conditional: *under* the SEU + softmax model, α measures sensitivity to the EU ranking the model attributes to the agent. The §7 application makes this concrete — the licensed reading is the within-design *comparative* one (§7.6.3), not an absolute-rationality ranking. Model misspecification is a substantive interpretive risk for any procedural-evaluation instrument and is acknowledged here rather than buried.

8.5 Limitations and extensions

Precise δ estimation. The modest finite-sample δ gain (§6.4.2) is a limitation, not a barrier in principle. The available levers — a δ -information-optimal lottery design (e.g., triangulating contrasts that pit the certain intermediate consequence against a 50/50 mix of the extremes), substantially larger samples, and prior regularization on δ — are *not independent*: a δ -optimal design helps less when α is small (shallow softmax at moderate α is a co-dampener), so precise δ recovery likely requires improvements on *multiple* axes at once. Naming the levers separately is a useful taxonomy, not a single-lever roadmap. A δ -optimal lottery-design study is named here as future work, not promised as a deliverable of this paper.

The single- α assumption. A single α governing both the uncertain and risky blocks (§5.3) is substantive and testable. It is essential to the §6.4 quantity-versus-type reading, which is why that reading is stated as explicitly conditional on it. The natural relaxation is a block-specific model fitting separate α_{unc} , α_{risky} (our internal `m_2`); it is named here, not pursued.

Menu size and sensitivity. The model places no restriction on the number of alternatives in a decision problem: the available set $\mathcal{A}_m$ may be of any size, the softmax of §2.2 is taken over it, and the Stan implementation already accepts a per-problem alternative count (Appendix C). The classical prospect-theory paradigm and its large-scale replication (Kahneman and Tversky 1979; Tversky and Kahneman 1992; Ruggeri et al. 2020) restricted attention to pairwise choice, appropriately for their purpose; the present formulation does not, which makes the *relationship between sensitivity and menu size* an estimable object rather than a fixed feature of the design. Treating menu size N_m as a covariate on $\log \alpha$ — e.g. $\log \alpha = \gamma_0 + \gamma_1 N_m$ in the hierarchical vehicle above — would let one ask whether sensitivity rises or falls as alternatives are added; one illustrative, model-agnostic hypothesis is that a larger menu raises the cognitive load on the decision maker and thereby lowers α , but the sign is left open. This is named, not pursued: the §7 designs randomize and position-counterbalance N_m and fit a single α pooled across menu sizes, so estimating a menu-size effect cleanly would require a purpose-built design that varies menu size as a factor rather than a nuisance.

Functional form. Linear-softmax belief formation ($\psi_r = \text{softmax}(\beta w_r)$) is a strong functional form; extensions to nonlinear belief formation and richer feature maps are possible.

Hierarchical extensions. The §7 application motivates the hierarchical extension `h_m01` to address the design-induced cross-condition structure of §7.7: because the five conditions reuse a single fixed pool of alternatives (with the LLM assessments and embeddings regenerated independently

at each temperature), fitting `m_01` independently per condition leaves two things on the table — partial pooling of information across conditions (generalization beyond the realized pool) and an explicit pool-by-temperature interaction. Modelling $\log \alpha_c = \gamma_0 + \gamma_1 T_c + \eta_c$ turns between-condition contrasts into estimated regression effects on $\log \alpha$ and lets the shared pool be modelled as a common random effect rather than ignored. `h_m01` will be the analysis vehicle of a planned alignment-study companion paper and is out of scope here.

Further application studies. A tier-stratified Ellsberg analysis, a non-SEU comparator instrument (e.g., a small α -MEU model paired with the SEU instrument to triangulate ambiguity-driven versus EU-driven choice), and the alignment study itself are pursued in companion work and are out of scope here.

8.6 Closing

A precise definition of sensitivity, a formal identifiability analysis, a computational validation pipeline, and an end-to-end illustrative application together provide a transparent, reusable framework for *procedurally* evaluating decision makers — human or machine — against a stated rationality standard expressible as expected-value maximization, with SEU as the canonical case. The framework reports where a parameter is identifiable but not precisely estimable, where in-principle arguments fail to predict finite-sample behavior, where marginal calibration is blind to a jointly weakly-informed posterior, and where an instrument declines to support an effect its own diagnostics cannot resolve. Reporting these limits is part of what makes the framework usable as a measurement instrument rather than a procedure that returns an effect by construction.

Appendix A. Proofs of the Softmax-Choice Properties

Proofs of the three properties stated in §2.3, plus the scale-invariance result underlying the $\mathbf{v}$ -endpoint convention of §2.6. These are the only results in the paper that hold as closed-form proofs without a genericity or curvature caveat.

Throughout, fix a decision problem m with available set $\mathcal{A} = \{r : I_{m,r} = 1\}$ and a value function V on it, write $P(r) = \exp(\alpha V(r))/Z(\alpha)$ with partition function $Z(\alpha) = \sum_{j \in \mathcal{A}} \exp(\alpha V(j))$, $V^* = \max_{r \in \mathcal{A}} V(r)$, $\mathcal{R}^* = \{r \in \mathcal{A} : V(r) = V^*\}$, and $\mathcal{R}^- = \mathcal{A} \setminus \mathcal{R}^*$. All maxima, minima, and sums are taken over the available set $\mathcal{A}$.

A.1 Monotonicity in sensitivity (Property 1)

i Theorem A.1 (Monotonicity). Holding V fixed: (i) for any two alternatives $r, s \in \mathcal{A}$ the pairwise log-odds are linear in α , $\partial_\alpha \log[P(r)/P(s)] = V(r) - V(s)$, so the odds move monotonically toward the higher-valued alternative; (ii) the total probability $P(\mathcal{R}^*) = \sum_{r \in \mathcal{R}^*} P(r)$ on the value-maximizing set is strictly increasing in α ; (iii) each $r \in \mathcal{R}^*$ has $P(r)$ strictly increasing in α and each value-minimizing r (with $V(r) = \min_j V(j)$) has $P(r)$ strictly decreasing. A non-maximal alternative with $V(r) > \mathbb{E}_\alpha[V]$ has $\partial_\alpha P(r) > 0$, so individual non-maximal probabilities need not be monotone in α .

Proof. (i) From $P(r) = \exp(\alpha V(r))/Z(\alpha)$, $\log[P(r)/P(s)] = \alpha(V(r) - V(s))$, whose α -derivative is $V(r) - V(s)$.

(iii) Differentiating $P(r)$ by the quotient rule with $Z'(\alpha) = \sum_j V(j) \exp(\alpha V(j))$,

$$\frac{\partial P(r)}{\partial \alpha} = P(r) \left[V(r) - \frac{Z'(\alpha)}{Z(\alpha)} \right] = P(r) [V(r) - \mathbb{E}_\alpha[V]],$$

where $\mathbb{E}_\alpha[V] = \sum_j P(j) V(j)$. For any finite α every alternative receives positive probability, so $\min_j V(j) < \mathbb{E}_\alpha[V] < V^*$ strictly. Hence $\partial P(r)/\partial \alpha > 0$ for $r \in \mathcal{R}^*$ and $\partial P(r)/\partial \alpha < 0$ for any r attaining $\min_j V(j)$; for an intermediate r the sign equals that of $V(r) - \mathbb{E}_\alpha[V]$, which can be positive.

(ii) Writing $P^*(\alpha) = P(\mathcal{R}^*)$ and summing the derivative identity over $r \in \mathcal{R}^*$ (where $V(r) = V^*$),

$$\frac{\partial P^*}{\partial \alpha} = \sum_{r \in \mathcal{R}^*} P(r) [V^* - \mathbb{E}_\alpha[V]] = P^*(V^* - \mathbb{E}_\alpha[V]) > 0,$$

since $\mathbb{E}_\alpha[V] < V^*$ strictly. ■

A.2 Optimization limit (Property 2)

i Theorem A.2. As $\alpha \rightarrow \infty$, $P(r) \rightarrow 1/|\mathcal{R}^*|$ for $r \in \mathcal{R}^*$ and $P(r) \rightarrow 0$ for $r \in \mathcal{R}^-$.

Proof. Divide numerator and denominator of $P(r)$ by $\exp(\alpha V^*)$. For $r \in \mathcal{R}^*$,

$$P(r) = \frac{1}{|\mathcal{R}^*| + \sum_{j \in \mathcal{R}^-} \exp(\alpha[V(j) - V^*])} \rightarrow \frac{1}{|\mathcal{R}^*|},$$

since $V(j) - V^* < 0$ for $j \in \mathcal{R}^-$. For $r \in \mathcal{R}^-$ the numerator is $\exp(\alpha[V(r) - V^*]) \rightarrow 0$ while the denominator is bounded below by $|\mathcal{R}^*| > 0$, so $P(r) \rightarrow 0$. ■

Remark. Ties ($|\mathcal{R}^*| > 1$) have measure zero under the continuous priors of §4; the unique-maximizer case is generic and gives deterministic optimal choice in the limit.

A.3 Uniform-choice limit (Property 3)

i Theorem A.3. As $\alpha \rightarrow 0$, $P(r) \rightarrow 1/|\mathcal{R}|$ for all $r \in \mathcal{R}$, independent of V .

Proof. By the expansion $\exp(x) = 1 + x + O(x^2)$,

$$P(r) = \frac{1 + \alpha V(r) + O(\alpha^2)}{|\mathcal{R}| + \alpha \sum_j V(j) + O(\alpha^2)} \rightarrow \frac{1}{|\mathcal{R}|}$$

as $\alpha \rightarrow 0$. Equivalently, $\log P(r) = -\log |\mathcal{R}| + \alpha[V(r) - \bar{V}] + O(\alpha^2)$ with $\bar{V} = |\mathcal{R}|^{-1} \sum_j V(j)$, so $P(r) \rightarrow 1/|\mathcal{R}|$. ■

A.4 Scale invariance and the utility-endpoint convention

i Theorem A.4 (Scale invariance). Let $\tilde{v}_k = a v_k + b$ with $a > 0$. Then $\tilde{\eta}_r = a \eta_r + b$ for all r , and

$$P(\text{choose } r \mid \alpha, \tilde{\mathbf{v}}) = P(\text{choose } r \mid \alpha a, \mathbf{v}).$$

The pairs $(\alpha, \tilde{\mathbf{v}})$ and $(\alpha a, \mathbf{v})$ generate identical choice probabilities.

Proof. Since $\sum_k \psi_{r,k} = 1$, $\tilde{\eta}_r = \sum_k \psi_{r,k}(a v_k + b) = a \eta_r + b$. Substituting into softmax, the additive constant αb cancels between numerator and denominator and the multiplicative factor combines with α :

$$P(r \mid \alpha, \tilde{\mathbf{v}}) = \frac{\exp(\alpha[a\eta_r + b])}{\sum_j \exp(\alpha[a\eta_j + b])} = \frac{\exp(\alpha a \eta_r)}{\sum_j \exp(\alpha a \eta_j)} = P(r \mid \alpha a, \mathbf{v}). \quad \blacksquare$$

Consequence. Without fixing the utility scale, α and the scale of utility are confounded: scaling utilities by a is equivalent to scaling sensitivity by $1/a$. Fixing the endpoints $v_1 = 0$, $v_K = 1$ (the $\mathbf{v}$ -endpoint convention of §2.6) selects a representative from the affine equivalence class and renders α identifiable, with the precise interpretation that $\log[P(r)/P(s)] = \alpha(\eta_r - \eta_s)$ on a standardized scale where the full utility range spans one unit.

Appendix B. Identifiability Results

B.1 and B.3 are proofs under explicit genericity conditions; B.4 records what uncertain choices do and do not pin down about β , identifying it as an irreducible nuisance; B.2 is not a proposition but a brief empirical note on the Bayesian workflow, recording that uncertain-choice data weakly inform (β, δ) as seen directly in parameter recovery.

B.1 Identifiability of α from η

i Proposition B.1. Let $\eta = (\eta_r)_{r \in \mathcal{R}}$ be fixed. Suppose there is a menu $\mathcal{M} \subseteq \mathcal{R}$ of positive design probability and two alternatives $r, s \in \mathcal{M}$ with $\eta_r \neq \eta_s$. Then the map $\alpha \mapsto P(\cdot \mid \alpha, \eta)$ restricted to $\mathcal{M}$ is injective on $(0, \infty)$; hence α is globally identifiable from choice probabilities on $\mathcal{M}$.

Proof. On $\mathcal{M}$ the softmax (Equation 1) gives, for the pair r, s ,

$$g(\alpha) := \log \frac{P(r \mid \alpha, \eta)}{P(s \mid \alpha, \eta)} = \alpha(\eta_r - \eta_s).$$

Since $\eta_r - \eta_s \neq 0$, g is strictly monotone (linear with nonzero slope) in α , hence injective: distinct α values induce distinct log-odds, therefore distinct choice-probability vectors on $\mathcal{M}$. As the menu has positive design probability, the choice distribution determines $g(\alpha)$ and thus α . $\blacksquare$

Remark (genericity). The condition “not all η_r equal on some positive-probability menu” replaces the vague “sufficient feature diversity”: it is exactly the condition under which softmax curvature in α is non-trivial. It fails only in the measure-zero event that every presented menu has perfectly tied expected utilities.

B.2 Weakly-informed (β, δ) in $\mathbf{m}_0$: a workflow note

This is not a proposition but a record of what the Bayesian workflow shows about (β, δ) in $\mathbf{m}_0$. The structural reason they are hard to separate is that expected utility composes beliefs and utilities multiplicatively,

$$\eta_r = \text{softmax}(\beta w_r)^\top \mathbf{v}(\delta),$$

so uncertain choices see (β, δ) only through the scalar η_r . A change in beliefs offset by a compensating change in utilities leaves the implied expected utilities — and hence the choice probabilities — almost unchanged. The $\mathbf{v}$ -endpoint convention (§2.6) fixes the zero and unit of the utility scale, but the multiplicative (β, δ) coupling is not an indeterminacy that any convention removes.

What recovery shows. Parameter recovery in $\mathbf{m}_0$ (§4.3) returns wide marginal credible intervals for both β and δ — in sharp contrast to the tight, well-calibrated α intervals — together with *correlated β - δ estimation errors* across recovery replicates: the posterior concentrates on a trade-off between beliefs and utilities rather than on either separately. That correlated, barely-contracted joint posterior is the practical signature of the multiplicative coupling and is what we mean by calling (β, δ) *weakly informed* by uncertain choices. No strict non-identifiability is claimed, and no gauge group on (β, δ) beyond the β row-shift of §2.6 is asserted; the point is that uncertain-choice data alone supply little information for separating beliefs from utilities. The matched-design study of §6.4 quantifies how little the risky block adds to this separation at realistic n .

B.3 Global identifiability of δ in $\mathbf{m}_1$

i Proposition B.3. Let the risky problems present lotteries $\{\pi_s\}_{s=1}^S \subset \Delta^{K-1}$ whose differences span the $(K-1)$ -dimensional tangent space of the simplex — equivalently, the set contains an affinely independent subset of K lotteries (which requires $S \geq K$ and a generic configuration) — and let the menu-contrast graph be connected. Then, with $\alpha > 0$ known to be positive and the endpoint convention $v_1 = 0, v_K = 1$ in force, the utility vector $\mathbf{v}$ — and hence δ — together with α is globally identified from risky-choice probabilities alone.

Proof. For each risky menu and each co-menu pair s, s' , the softmax (Equation 1) gives the log-odds

$$\log \frac{P(s)}{P(s')} = \alpha (\pi_s - \pi_{s'})^\top \mathbf{v}.$$

The choice probabilities therefore identify the products $c_s := \alpha (\pi_s - \pi_1)^\top \mathbf{v}$ for every lottery s sharing a menu with a common reference π_1 , and — chaining across the connected menu-contrast graph — for all $s = 2, \dots, S$. We do *not* invoke Proposition B.1 here: α and $\mathbf{v}$ are identified jointly, as follows. Ratios $c_s/c_{s'}$ cancel the unknown $\alpha > 0$ and identify the linear functionals $(\pi_s - \pi_1)^\top \mathbf{v}$ up to one common positive scale. Because the differences $\{\pi_s - \pi_1\}$ span the $(K-1)$ -dimensional tangent space, these functionals determine $\mathbf{v}$ up to a positive affine transformation; the endpoint convention $v_1 = 0, v_K = 1$ fixes that transformation, so $\mathbf{v}$ is identified uniquely. Finally $\alpha = c_s/(\pi_s - \pi_1)^\top \mathbf{v}$ for any s with non-zero denominator (one exists, as $\mathbf{v}$ is non-constant), identifying α . The cumulative-sum map $\delta \mapsto \mathbf{v}$ being a bijection onto the ordered-utility set, δ is identified. ■

Remark. The substantive hypothesis is a design condition the experimenter controls: the presented lotteries' differences must span the simplex tangent space (equivalently, the design contains an affinely independent subset of K lotteries), and the menus must be linked into a connected contrast

graph. Affine independence of *all* S lotteries is neither required nor possible when $S > K$ (here $S = 15$, $K = 3$); what is needed is a spanning affinely independent *subset*.

B.4 Uncertain choices sharpen utilities; β remains a nuisance

i Proposition B.4. Suppose δ (hence $\mathbf{v}$) is identified by B.3 and $\alpha > 0$ by B.1. Then the uncertain-choice probabilities identify the expected utilities $\eta_r = \boldsymbol{\psi}_r^\top \mathbf{v}$ up to a per-menu additive constant, and constrain β only through the composite map $\beta \mapsto (\text{softmax}(\beta w_r)^\top \mathbf{v})_r$. For $K \geq 3$ a scalar expected utility does not determine the $(K - 1)$ -dimensional belief simplex $\boldsymbol{\psi}_r$, so absent an independent rank/Jacobian condition on this composite map, β is identified at most up to the level sets of $\beta \mapsto (\eta_r)_r$ together with the additive row-shift gauge of §2.6. In the uncertain-choice model β is thus an irreducible nuisance; the contribution of the risky block (B.3) is to sharpen $\mathbf{v}$, not to recover β .

Proof. With α and $\mathbf{v}$ fixed, Proposition B.1 applied within each menu shows the choice probabilities determine the within-menu contrasts of $\eta_r = \boldsymbol{\psi}_r^\top \mathbf{v}$, hence the η_r up to a common per-menu constant. Each such value is a *single* linear equation in the $(K - 1)$ free coordinates of $\boldsymbol{\psi}_r$; for $K \geq 3$ the level set $\{\boldsymbol{\psi} : \boldsymbol{\psi}^\top \mathbf{v} = \eta_r\}$ is $(K - 2)$ -dimensional whenever $\mathbf{v}$ is non-constant, so $\boldsymbol{\psi}_r$ is not recovered. The map $\boldsymbol{\psi} \mapsto \boldsymbol{\psi}^\top \mathbf{v}$ is therefore not injective, and β enters the likelihood only through the composite $\beta \mapsto (\eta_r)_r$. Any two weight matrices inducing the same expected utilities on every presented menu — including those related by the row-shift gauge, for which $\text{softmax}(\beta w_r)$ itself is unchanged — are observationally indistinguishable. Point identification of β would require the composite map to be injective modulo the gauge under the design, an additional rank condition the scalar expected utilities do not supply. ■

Appendix C. Stan Model Listings

*Full listings of the Stan programs referenced by the brief code excerpts in §4 and §6. Two model pairs are used throughout. For each model there is an **inference** program (`.stan`) that takes observed choices as data and samples the posterior, and a matched **simulation** program (`_sim.stan`) that draws parameters from the prior and generates synthetic choices in **generated quantities** — the data-generating half of every prior-predictive, parameter-recovery, and SBC run. The calibrated-prior variant `m_01` used in the application (§7) is structurally identical to `m_0` and is given as a one-line prior diff. Listings reflect the pinned commit of Appendix E.0; the canonical files live at `models/` in the repository.*

The notation matches §2.6: `alpha` is the sensitivity α , `beta` the belief-feature matrix β , `delta` the utility increments δ (a **simplex**, so `upsilon = v` is ordered with the endpoint convention $v_1 = 0$ built in via `cumulative_sum(append_row(0, delta))`), `psi` the per-alternative belief simplices $\boldsymbol{\psi}_r$, `eta` the expected utilities η_r , and `chi` the per-problem choice probabilities. The `I` (uncertain) and `J` (risky) indicator arrays encode which alternatives appear in each problem; choices are `y` (uncertain) and `z` (risky).

C.1 `m_0.stan` — uncertain-choice-only inference model


```

data{
  int<lower=1> M; // the number of decision problems
  int<lower=2> K; // the number of possible consequences
  int<lower=1> D; // the number of dimensions to describe an alternative
  int<lower=2> R; // the number of distinct alternatives
  array[R] vector[D] w; // the descriptions of each distinct alternative
  array[M,R] int<lower=0,upper=1> I; // I[m,r] = 1 if alternative r is in problem m
  array[M] int<lower=1> y; // selected alternative (bounds checked in transformed data)
}

transformed data {
  array[M] int<lower=2> N; // the number of alternatives in each decision problem
  int total_alternatives = 0;
  for (m in 1:M) {
    N[m] = sum(I[m]);
    total_alternatives += N[m];
    if (y[m] > N[m]) reject("y[" , m, "] = ", y[m], " must be <= N[" , m, "] = ", N[m]);
  }
  // Construct x from w and I
  array[total_alternatives] vector[D] x;
  {
    int pos = 1;
    for (m in 1:M) {
      for (r in 1:R) {
        if (I[m, r] == 1) { x[pos] = w[r]; pos += 1; }
      }
    }
  }
}

parameters{
  real<lower=0> alpha; // sensitivity parameter (decision noise); strictly positive
  matrix[K,D] beta; // coefficients mapping descriptions to subjective probabilities
  simplex[K - 1] delta; // utility differences on unit scale (ensures ordered utilities)
}

transformed parameters{
  array[sum(N)] simplex[K] psi; // subjective probability over outcomes per alternative
  ordered[K] epsilon; // the subjective utility of each consequence
  vector[sum(N)] eta; // expected utility of each alternative on unit scale
  array[M] simplex[max(N)] chi; // choice probs with padding
  for(i in 1:sum(N)){ psi[i] = softmax(beta*x[i]); }
  epsilon = cumulative_sum(append_row(0, delta));
  for(i in 1:sum(N)){ eta[i] = dot_product(psi[i],epsilon); }
  {
    int pos = 1;
    for(i in 1:M){
      vector[N[i]] problem_eta = segment(eta, pos, N[i]);
      chi[i] = append_row(softmax(alpha * problem_eta), rep_vector(0, max(N) - N[i]));
      pos += N[i];
    }
  }
}

```

```

    }
  }
}
model{
  alpha ~ lognormal(0, 1);           // prior on choice sensitivity
  to_vector(beta) ~ std_normal();    // prior on belief-feature coefficients
  delta ~ dirichlet(rep_vector(1,K-1)); // prior on ordered unit-scale utility increments
  for(i in 1:M){ y[i] ~ categorical(chi[i]); }
}
generated quantities {
  vector[M] log_lik;
  for (i in 1:M) { log_lik[i] = categorical_lpmf(y[i] | chi[i]); }
  array[M] int y_pred;
  for (i in 1:M) { y_pred[i] = categorical_rng(chi[i]); }

  // Posterior-predictive check statistics (one-sided Bayesian p-values):
  // log-likelihood discrepancy, modal-choice accuracy, and sum of chosen
  // probabilities, each comparing T(y_pred) to T(y_obs).
  real T_obs_ll = sum(log_lik);
  real T_rep_ll = 0;
  for (m in 1:M) { T_rep_ll += categorical_lpmf(y_pred[m] | chi[m]); }
  int<lower=0,upper=1> ppc_ll = (T_rep_ll >= T_obs_ll) ? 1 : 0;

  int T_obs_modal = 0; int T_rep_modal = 0;
  for (m in 1:M) {
    real max_prob = chi[m][1];
    for (j in 2:N[m]) { if (chi[m][j] > max_prob) max_prob = chi[m][j]; }
    T_obs_modal += (chi[m][y[m]] >= max_prob - 1e-9) ? 1 : 0;
    T_rep_modal += (chi[m][y_pred[m]] >= max_prob - 1e-9) ? 1 : 0;
  }
  int<lower=0,upper=1> ppc_modal = (T_rep_modal >= T_obs_modal) ? 1 : 0;

  real T_obs_prob = 0; real T_rep_prob = 0;
  for (m in 1:M) { T_obs_prob += chi[m][y[m]]; T_rep_prob += chi[m][y_pred[m]]; }
  int<lower=0,upper=1> ppc_prob = (T_rep_prob >= T_obs_prob) ? 1 : 0;
}

```

C.2 m_0_sim.stan — uncertain-choice data generator

The simulation program shares `m_0`'s `data/transformed data` shape but draws (α, β, δ) from the prior inside `generated quantities` (with the α -prior hyperparameters passed as data, so a single program serves both the `Lognormal(0,1)` default and any calibrated prior), simulates choices `y`, and reports `total_seu_max_selected` — the count of problems in which the sampled choice coincided with the SEU-maximizing alternative, the prior-predictive diagnostic of §4.2.

```

data {
  int<lower=1> M; int<lower=2> K; int<lower=1> D; int<lower=2> R;
  array[R] vector[D] w;
}

```

```

array[M,R] int<lower=0,upper=1> I;
real<lower=0> alpha_mean; // mean for log(alpha) (default 0)
real<lower=0> alpha_sd;    // sd for log(alpha) (default 1)
real<lower=0> beta_sd;     // sd for beta coefficients (default 1)
}
transformed data {
  array[M] int<lower=2> N; int total_alts = 0;
  for (m in 1:M) { N[m] = sum(I[m]); total_alts += N[m]; }
  array[total_alts] vector[D] x;
  {
    int pos = 1;
    for (m in 1:M) for (r in 1:R) if (I[m, r] == 1) { x[pos] = w[r]; pos += 1; }
  }
}
generated quantities {
  real alpha = lognormal_rng(alpha_mean, alpha_sd);
  matrix[K,D] beta;
  for (k in 1:K) for (d in 1:D) beta[k,d] = normal_rng(0, beta_sd);
  simplex[K-1] delta = dirichlet_rng(rep_vector(1.0, K-1));

  array[total_alts] vector[K] psi;
  for (i in 1:total_alts) psi[i] = softmax(beta * x[i]);
  vector[K] upsiilon = cumulative_sum(append_row(0, delta));
  vector[total_alts] eta;
  for (i in 1:total_alts) eta[i] = dot_product(to_vector(psi[i]), upsiilon);

  array[M] int y;
  array[M] int<lower=0,upper=1> selected_seu_max;
  {
    int pos = 1;
    for (i in 1:M) {
      vector[N[i]] problem_eta = segment(eta, pos, N[i]);
      vector[N[i]] choice_probs = softmax(alpha * problem_eta);
      y[i] = categorical_rng(choice_probs);
      real max_eta = max(problem_eta);
      selected_seu_max[i] = (abs(problem_eta[y[i]] - max_eta) < 1e-10) ? 1 : 0;
      pos += N[i];
    }
  }
  int<lower=0,upper=M> total_seu_max_selected = sum(selected_seu_max);
}

```

(The full file additionally stores the per-problem *problem_etas* and *choice_probabilities* for inspection; those bookkeeping arrays are elided here.)

C.3 m_1.stan — combined uncertain + risky inference model

m_1 adds a risky block: a second set of N problems over S lotteries with *objective* probability simplices x supplied as data. The lotteries enter expected utility directly ($\text{eta_risky} = \text{dot_product}(x_risky, \text{upsilon})$), with no β — this β -free channel is what identifies δ in principle (§5, Proposition B.3). Crucially, α and the utility vector upsilon are **shared** across the two blocks.

```
data {
  // Uncertain problems
  int<lower=1> M; int<lower=2> K; int<lower=1> D; int<lower=2> R;
  array[R] vector[D] w;
  array[M,R] int<lower=0,upper=1> I;
  array[M] int<lower=1> y;
  // Risky problems
  int<lower=1> N; int<lower=2> S;
  array[S] simplex[K] x;           // objective lottery simplices
  array[N,S] int<lower=0,upper=1> J;
  array[N] int<lower=1> z;
}
transformed data {
  array[M] int<lower=2> N_uncertain; int total_uncertain_alts = 0;
  for (m in 1:M) {
    N_uncertain[m] = sum(I[m]); total_uncertain_alts += N_uncertain[m];
    if (y[m] > N_uncertain[m]) reject("y[, m, "] out of range");
  }
  array[N] int<lower=2> N_risky; int total_risky_alts = 0;
  for (n in 1:N) {
    N_risky[n] = sum(J[n]); total_risky_alts += N_risky[n];
    if (z[n] > N_risky[n]) reject("z[, n, "] out of range");
  }
  array[total_uncertain_alts] vector[D] x_uncertain;
  {
    int pos = 1;
    for (m in 1:M) for (r in 1:R) if (I[m,r]==1) { x_uncertain[pos]=w[r]; pos+=1; }
  }
  array[total_risky_alts] simplex[K] x_risky;
  {
    int pos = 1;
    for (n in 1:N) for (s in 1:S) if (J[n,s]==1) { x_risky[pos]=x[s]; pos+=1; }
  }
}
parameters {
  real<lower=0> alpha;    // shared sensitivity
  matrix[K,D] beta;      // belief features (uncertain block only)
  simplex[K-1] delta;    // shared utility increments
}
transformed parameters {
  ordered[K] upsilon = cumulative_sum(append_row(0, delta)); // shared utilities
```

```

// Uncertain block
array[total_uncertain_alts] simplex[K] psi;
vector[total_uncertain_alts] eta_uncertain;
array[M] simplex[max(N_uncertain)] chi_uncertain;
for (i in 1:total_uncertain_alts) psi[i] = softmax(beta * x_uncertain[i]);
for (i in 1:total_uncertain_alts) eta_uncertain[i] = dot_product(psi[i], upsi);
{
  int pos = 1;
  for (m in 1:M) {
    vector[N_uncertain[m]] pe = segment(eta_uncertain, pos, N_uncertain[m]);
    chi_uncertain[m] = append_row(softmax(alpha*pe),
                                rep_vector(0, max(N_uncertain)-N_uncertain[m]));
    pos += N_uncertain[m];
  }
}

// Risky block (beta-free: lotteries enter EU directly)
vector[total_risky_alts] eta_risky;
array[N] simplex[max(N_risky)] chi_risky;
for (i in 1:total_risky_alts) eta_risky[i] = dot_product(x_risky[i], upsi);
{
  int pos = 1;
  for (n in 1:N) {
    vector[N_risky[n]] pe = segment(eta_risky, pos, N_risky[n]);
    chi_risky[n] = append_row(softmax(alpha*pe),
                             rep_vector(0, max(N_risky)-N_risky[n]));
    pos += N_risky[n];
  }
}
}

model {
  alpha ~ lognormal(0, 1);
  to_vector(beta) ~ std_normal();
  delta ~ dirichlet(rep_vector(1, K-1));
  for (m in 1:M) y[m] ~ categorical(chi_uncertain[m]);
  for (n in 1:N) z[n] ~ categorical(chi_risky[n]);
}

generated quantities {
  vector[M] log_lik_uncertain;
  vector[N] log_lik_risky;
  for (m in 1:M) log_lik_uncertain[m] = categorical_lpmf(y[m] | chi_uncertain[m]);
  for (n in 1:N) log_lik_risky[n] = categorical_lpmf(z[n] | chi_risky[n]);
  real log_lik_total = sum(log_lik_uncertain) + sum(log_lik_risky);
  array[M] int y_pred; array[N] int z_pred;
  for (m in 1:M) y_pred[m] = categorical_rng(chi_uncertain[m]);
  for (n in 1:N) z_pred[n] = categorical_rng(chi_risky[n]);
  // Block-wise and combined PPC statistics (ppc_ll / ppc_modal / ppc_prob,

```

```
// built exactly as in m_0.stan for each block) are computed here.
}
```

C.4 m_1_sim.stan — combined data generator

The matched simulator draws shared (α, δ) and uncertain-only β from the prior, then generates both y and z . The four matched-design conditions of §6.2 (Appendix D.4) are produced by slicing the choices this program generates — the same simulated parameters feed every condition, which is what makes the A/B/C/D comparison a *matched* one.

```
data {
  int<lower=1> M; int<lower=2> K; int<lower=1> D; int<lower=2> R;
  array[R] vector[D] w; array[M,R] int<lower=0,upper=1> I;
  int<lower=1> N; int<lower=2> S;
  array[S] simplex[K] x; array[N,S] int<lower=0,upper=1> J;
  real<lower=0> alpha_mean; real<lower=0> alpha_sd; real<lower=0> beta_sd;
}

transformed data {
  array[M] int<lower=2> N_uncertain; int total_uncertain_alts = 0;
  for (m in 1:M) { N_uncertain[m] = sum(I[m]); total_uncertain_alts += N_uncertain[m]; }
  array[total_uncertain_alts] vector[D] x_uncertain;
  { int pos=1; for (m in 1:M) for (r in 1:R) if (I[m,r]==1){x_uncertain[pos]=w[r];pos+=1;} }
  array[N] int<lower=2> N_risky; int total_risky_alts = 0;
  for (n in 1:N) { N_risky[n] = sum(J[n]); total_risky_alts += N_risky[n]; }
  array[total_risky_alts] simplex[K] x_risky;
  { int pos=1; for (n in 1:N) for (s in 1:S) if (J[n,s]==1){x_risky[pos]=x[s];pos+=1;} }
}

generated quantities {
  real alpha = lognormal_rng(alpha_mean, alpha_sd);
  matrix[K,D] beta;
  for (k in 1:K) for (d in 1:D) beta[k,d] = normal_rng(0, beta_sd);
  simplex[K-1] delta = dirichlet_rng(rep_vector(1.0, K-1));
  vector[K] upsi = cumulative_sum(append_row(0, delta));

  // Uncertain choices
  array[total_uncertain_alts] vector[K] psi;
  for (i in 1:total_uncertain_alts) psi[i] = softmax(beta * x_uncertain[i]);
  vector[total_uncertain_alts] eta_uncertain;
  for (i in 1:total_uncertain_alts) eta_uncertain[i] = dot_product(to_vector(psi[i]), upsi);
  array[M] int y; array[M] int<lower=0,upper=1> selected_seu_max_uncertain;
  {
    int pos = 1;
    for (m in 1:M) {
      vector[N_uncertain[m]] pe = segment(eta_uncertain, pos, N_uncertain[m]);
      vector[N_uncertain[m]] cp = softmax(alpha * pe);
      y[m] = categorical_rng(cp);
      selected_seu_max_uncertain[m] = (abs(pe[y[m]] - max(pe)) < 1e-10) ? 1 : 0;
      pos += N_uncertain[m];
    }
  }
}
```

```

    }
  }

  // Risky choices (beta-free)
  vector[total_risky_alts] eta_risky;
  for (i in 1:total_risky_alts) eta_risky[i] = dot_product(to_vector(x_risky[i]), epsilon);
  array[N] int z; array[N] int<lower=0,upper=1> selected_seu_max_risky;
  {
    int pos = 1;
    for (n in 1:N) {
      vector[N_risky[n]] pe = segment(eta_risky, pos, N_risky[n]);
      vector[N_risky[n]] cp = softmax(alpha * pe);
      z[n] = categorical_rng(cp);
      selected_seu_max_risky[n] = (abs(pe[z[n]] - max(pe)) < 1e-10) ? 1 : 0;
      pos += N_risky[n];
    }
  }
  int total_seu_max_selected =
    sum(selected_seu_max_uncertain) + sum(selected_seu_max_risky);
}

```

C.5 m_01.stan — calibrated-prior variant (application)

The application model `m_01` is **structurally identical** to `m_0` (§C.1) — same data, parameters, transformed parameters, and generated quantities. It differs only in the α prior, which is calibrated by the prior-predictive SEU-max-rate procedure of §7.3 (claim C8) rather than left at the convenience default. The single changed line in the `model` block is:

```

// m_0.stan:
alpha ~ lognormal(0, 1);      // convenience default
// m_01.stan:
alpha ~ lognormal(3.0, 0.75); // calibrated: median ~ 20, 90% CI ~ [5.5, 67],
                               // prior-implied SEU-max rate ~ 0.78 (insurance, K=3)

```

For the $K = 4$ Ellsberg cells the same SEU-max-rate calibration is re-run to set the α -prior hyperparameters at that menu size (Appendix D.5); the model code is otherwise unchanged.

Appendix D. Computational Details and Design Constants

*This appendix is the **canonical** location for the study-design constants and sampler settings used throughout the paper; the body (§§4, 6, 7) references D.0 and D.5 rather than re-listing. D.0 fixes the foundational (methodological) design; D.5 the application scale. D.1–D.4 give software versions, sampler configuration, the SBC protocol, and the matched-design comparison. Exact, machine-readable values live in `configs/*.json` and `applications/*/configs/` at the pinned commit (Appendix E.0); the constants below are the human-readable transcription.*

D.0 Foundational study-design constants

The methodological models `m_0` and `m_1` (§§4, 6) use a single small design, chosen to be representative rather than power-optimized:

Constant	Symbol	Value
Consequences per problem	K	3
Belief-feature dimension	D	5
Distinct uncertain alternatives	R	15
Distinct risky lotteries (<code>m_1</code>)	S	15
Uncertain problems	M	$\in \{25, 50\}$
Risky problems (<code>m_1</code>)	N	$\in \{25, 50\}$
Alternatives per problem	—	2–5 (uniform)
Feature distribution	—	$\mathcal{N}(0, 1)$, i.i.d.
Risky lottery simplices	π_s	fixed across iterations

The matched-design recovery study (D.4) instantiates these at the four (M, N) slices of the §6.2 table. The simulation-based-calibration protocol (D.3) uses $N_{\text{sbc}} = 999$ draws, thinning 4, a single chain, 20 rank-histogram bins, and the $L = N_{\text{sbc}}$ rank convention.

D.1 Software versions

All fits use **CmdStan v2.37.0** driven by **CmdStanPy** (≥ 1.2) under **Python 3.10** (conda environment `seu-sensitivity`). Numerical and post-processing support: NumPy (≥ 1.20), SciPy (≥ 1.7), pandas (≥ 1.3), scikit-learn (≥ 1.0 , used for the $D = 32$ PCA feature reduction of the insurance application, D.5), and Matplotlib (≥ 3.4). LLM choice data were collected via the OpenAI (≥ 1.0) and Anthropic (≥ 0.18) Python clients. The exact, pinned versions are recorded in `requirements.txt` and `environment.yml` at the commit of Appendix E.0; the versions above are the essential ones for reproducing the inference.

D.2 Sampler configuration and diagnostics policy

Sampler. Posterior fits use **4 chains** and **1000 sampling** iterations per chain (**4000 retained draws**), at CmdStan’s NUTS defaults — target acceptance `adapt_delta = 0.8` and maximum tree depth 10, not overridden. Warmup is 500 iterations per chain in the parameter-recovery driver and 1000 in the §7 application fits; either way 4000 post-warmup draws are retained, the denominator behind the §7.4 divergence and $\hat{R}$ -exceedance fractions. (The SBC fits of D.3 use a different, single-chain configuration.) Random seeds are deterministic per iteration: the simulation draw uses seed $12345 + i$ and the inference fit uses $54321 + i$ for iteration i , so every recovery iteration is independently reproducible.

Diagnostics policy. A fit is considered well-behaved when, for the key parameters, the potential-scale-reduction factor satisfies $\hat{R} \leq 1.01$ and both bulk- and tail-effective sample size are adequate for the reported quantiles; divergent transitions are tracked per fit and reported as a fraction. In the application (§7.4, claim C14) α — the only parameter of primary interest for the temperature analysis — is never $\hat{R}$ -flagged and has satisfactory ESS in all 20 fits; divergences are $\leq 0.15\%$ per fit; the only $\hat{R} > 1.01$ exceedances fall on the weakly-informed β, δ and the per-trial nuisance latents (η, ψ, v) in the hardest $K = 4$ fits — consistent with the §3.4 weakly-informed (β, δ) finding rather than a defect in the α inference.

D.3 SBC configuration and seed convention

Simulation-based calibration (§4.4, §6.5, claim C5) draws $N_{\text{sbc}} = 999$ prior samples from the matched `_sim` program (Appendix C.2, C.4), fits the inference model to each synthetic dataset, and ranks each true parameter within its thinned posterior draws. Thinning is 4, posteriors are run as a single chain, and the rank histograms use 20 bins under the $L = N_{\text{sbc}}$ convention so that uniform ranks span $\{0, \dots, N_{\text{sbc}}\}$. Calibration is read off both the rank histograms and the ECDF-difference plot with simultaneous KS bands (Modřák et al. 2023), which is more sensitive than the rank histogram alone. Each replicate i uses a single deterministic seed $123 + i$: the `_sbc` program draws the true parameters and simulates its synthetic dataset in the same pass it fits, so one seed governs both steps and each of the 999 replicates is reproducible in isolation. (The study design itself is generated once under NumPy seed 42 and held fixed across all replicates.)

D.4 The matched-design comparison (conditions A/B/C/D)

The matched study underlying §6.4 (claims C1–C3) is **not** a δ -optimal design study; it is a controlled slicing of one fixed design. Each of $n = 100$ iterations proceeds as:

1. generate a fresh study design *externally* — the feature draws w and the lottery menus, which enter the simulator as data $(w, I, x, J$; Appendix C.4), not as quantities `m_1_sim` itself draws — and draw a fresh true (α, β, δ) from the prior; each of the $n = 100$ iterations uses its own freshly generated design, not a single design held fixed across iterations;
2. with that design supplied as data, `m_1_sim` (Appendix C.4) draws choices *conditional on* the true parameters — uncertain choices y ($M = 50$) and risky choices z ($N = 50$);
3. fit four conditions to **slices of the same simulated choices**, so the true parameters are matched across conditions within the iteration:

Condition	Model	Uncertain M	Risky N	Total
A	<code>m_0</code>	25	—	25
B	<code>m_0</code>	50	—	50
C	<code>m_1</code>	25	25	50
D	<code>m_1</code>	50	50	100

The central comparison is **B vs C** (same total choice count; only the model and the *type* of choice differ) and the control is **A vs B** (more uncertain data alone). All reported effects are paired-iteration medians or bootstrap-over-iteration ratios with 90% CIs (10,000 resamples, design seed 20260617); see `spikes/report14_rerun_analysis.py`. The headline finding — that the β -free risky block yields no matched-count α -precision gain (C1) while improving- α comes from data *quantity* (A→B, C2), and the δ CI-width gain is $\approx 0.8\%$ (C3) — is a property of *this* matched slicing, not of a δ -optimized design.

D.5 Application-scale design constants

The 2×2 application (§7) uses the calibrated-prior model `m_01` (Appendix C.5) at a larger scale. Per LLM $\times$ task cell:

Constant	Insurance triage	Ellsberg urns
Consequences K	3	4
Base problems M	100	100
Position-counterbalanced presentations	$\times 3$	$\times 3$
Choices per temperature condition	300	300
Temperature conditions per cell	5	5
Alternatives per pool R	30	30
Belief-feature dimension D	32 (PCA)	— (objective urns)
Parameter-recovery iterations	20	20

Temperature grids differ by provider: GPT-4o uses $T \in \{0.0, 0.3, 0.7, 1.0, 1.5\}$; Claude 3.5 Sonnet uses $T \in \{0.0, 0.2, 0.5, 0.8, 1.0\}$ (the provider-specific upper bounds drive the unequal-grid caveat of §7.5.2a and the cross-LLM robustness check, claim C11).

Calibrated α priors. Insurance uses $\alpha \sim \text{Lognormal}(3.0, 0.75)$ (median ≈ 20 , 90% CI $\approx [5.5, 67]$, prior-implied SEU-max rate ≈ 0.78 ; claim C8). For the $K = 4$ Ellsberg cells the prior is re-calibrated by the same SEU-maximizer-rate prior-predictive procedure at that menu size, since the SEU-max rate at fixed α depends on K and the number of alternatives. Across all four cells the fit count is 5 temperatures $\times$ 4 cells = 20 posterior fits, plus $20 \times 4 = 80$ recovery iterations.

Appendix E. Reproducibility

Provenance, compute budget, and a per-figure manifest. The goal is that every key empirical claim in the paper can be regenerated from a pinned state of the supporting repository, with no number depending on an artifact that is not itself version-controlled.

E.0 Citation and provenance policy

Every key empirical claim cites the supporting repository at a **pinned commit or tag** — and, at the point of archival deposit, a DOI will be minted from that tagged commit (a Zenodo release) and substituted for the commit pin in the citation — never a homepage or blog URL. Supporting materials (data, analysis scripts, configs, figures) are **referenced at that pin, not copied** into the paper folder, so there is a single source of truth and no drift: the Stan listings of Appendix C, the design constants of Appendix D, and the figure manifest of E.2 all describe the repository *at the pin*. The dissemination pipeline (internal reports $\rightarrow$ homepage posts $\rightarrow$ this working paper) exists for reach; it is not a citation surface. The mapping from claim to artifact is maintained in the repository’s `claims_ledger.md`, in which every row cited in the paper carries status **computed** (finalized from a completed run) before it appears in body text.

The analysis is driven by a small set of generic scripts parameterized by JSON configs:

Script	Role
<code>analysis/prior_predictive.py</code>	prior-predictive SEU-max-rate checks (§4.2, §6.3, §7.3)
<code>analysis/parameter_recovery.py</code>	true-vs-estimated recovery (§4.3, §6.4, §7.4)
<code>analysis/sbc.py</code>	simulation-based calibration (§4.4, §6.5)
<code>analysis/posterior_predictive_checks.py</code>	PPC test statistics (§7.4)

Script	Role
<code>analysis/model_estimation.py</code>	posterior fits on observed LLM choices (§7.5)

driven by `configs/*.json` (methodological studies) and `applications/*/configs/` (the four LLM $\times$ task cells). The Stan programs are those of Appendix C under `models/`.

E.1 Compute budget

All models are small — a few hundred categorical choices, three structural parameters plus per-trial deterministic transforms — so a single posterior fit (4 chains $\times$ 1000 draws) completes in seconds to a low number of minutes on one core of a contemporary 8-core workstation. The end-to-end cost is dominated not by any single fit but by the **simulation-based-calibration** runs, which refit the model once per SBC draw: at $N_{\text{sbc}} = 999$ draws per model (D.3) these are the longest-running configs and account for the bulk of the wall-clock budget; the **matched-design recovery study** ($n = 100$ iterations $\times$ 4 conditions = 400 fits, D.4) and the **application fits** (20 posterior fits plus $20 \times 4 = 80$ recovery iterations, D.5) are each substantially cheaper. With the SBC runs parallelized across the 8 cores, a full end-to-end reproduction of every figure and table is an overnight-scale job rather than a multi-day one; the SBC configs are the ones to launch first and the only ones for which a single-core reproduction is uncomfortable. (These are order-of-magnitude figures from the development runs and are reported as such; a timed clean-room reproduction at the pinned commit will be the authoritative source and will be recorded alongside the archival deposit.)

E.2 Per-figure manifest

Each headline figure and table maps to a script, a config, and a seed, so it is reproducible in isolation. Every entry is *computed* — its numbers are finalized from a completed run. The foundational-report figures (§4.3, §4.4, §6.5) carry their canonical design inline rather than through a committed JSON config: the methodological reports ([04_parameter_recovery.qmd](#), [06_sbc_validation.qmd](#)) fix the canonical design constants of D.0 in-script and reuse the generic `parameter_recovery.py` / `sbc.py` drivers, so the committed `configs/*sbc*.json` and `configs/*recovery*.json` files (smaller smoke-scale designs) are not the pinned source for these three plots.

Table 9: Per-figure manifest: each headline figure or table mapped to its script, config, and seed.

Fig/Tab §	Description	Script	Config	Seed	Status
Fig 6.4	6.4.1 α/δ RMSE + CI-width, matched A/B/C/D ($n = 100$)	<code>spikes/report14_rerun_analysis.py</code>	<code>configs/m1_matched_recovery_n100_config.json</code>	design 20260617 / boot 20260617	computed
Tab (§7.5.5)	7.5 2×2 per-cell global-slope posteriors (C9–C13)	<code>analysis/model_estimation.py</code>	<code>applications/*/configs/</code>	per-cell (see cell data)	computed
Fig (§7.5.5)	7.5.5 2×2 forest plot of the per-cell global-slope posteriors (companion to Table 4)	<code>spikes/report_2x2_forest_spike.py</code>	canonical cell summaries (C9/C10/C12/C13)	deterministic (no RNG)	computed

Fig/Tab §	Description	Script	Config	Seed	Status
Fig 7.5.2	Claude-null MDE power curve; MDE ≈ 36 marker (C16)	<code>spikes/report16_-mde_spike.py</code>	inline (data <code>alpha_draws_T*.npz</code>)	20260618	computed
Fig (§4.3)	α true-vs-estimated recovery scatter (<code>m_0</code>)	<code>analysis/parameter_-recovery.py</code>	inline (<code>m_0</code> : $M25\ K3\ D5\ R15$; 50 iters, 4 chains $\times$ 1000)	design np 42; $12345 + i$ sim / $54321 + i$ fit (D.2)	computed
Fig (§4.4)	<code>m_0</code> SBC rank histograms + ECDF/KS	<code>analysis/sbc.py</code>	inline (<code>m_0</code> : $M25\ K3\ D5\ R15$; $N_{\text{sbc}}999$ thin4 1 chain)	design np 42; $123 + i$ per draw (D.3)	computed
Fig (§6.5)	<code>m_1</code> SBC rank histograms + ECDF/KS	<code>analysis/sbc.py</code>	inline (<code>m_1</code> : $M25\ N25\ K3\ D5\ R15\ S15$; $N_{\text{sbc}}999$ thin4 1 chain)	design np 42; $123 + i$ per draw (D.3)	computed

The cross-LLM robustness number (C11) and the per-cell diagnostics/PPC summaries (C14, C15) are reproduced by `spikes/report11_cross_llm_spike.py` and `spikes/report1415_diagnostics_ppc_spike.py` respectively (both seed 20260618), reading the committed per-condition posterior draws under `reports/applications/*/data/`. The 2×2 forest plot (§7.5.5) is rendered by `spikes/report_2x2_forest_spike.py` from the canonical report-level cell summaries (C9/C10/C12/C13), so it matches Table 4 by construction; the same script records a population-OLS cross-check over the committed draws in its results JSON (diagnostic only). It agrees with the canonical summaries on $P(\text{slope} < 0)$ and on both Ellsberg cells; it differs only on the insurance-cell slope *magnitude*, because the canonical 2×2 entry reports the **report-level** slope — an OLS through the four per-condition α posterior medians (≈ -3.6 for the Claude insurance cell) — whereas the diagnostic pools all posterior draws into a single **population OLS** (≈ -2.9 draw-level median). Claim C16 records this convention (report-level -3.6 as the canonical headline, draw-level ≈ 2.9 as the quantity the §7.5.2 MDE power calculation operates on), so the two numbers are the same effect read at two resolutions, not a discrepancy.

- Allais, Maurice. 1953. “Le Comportement de l’homme Rationnel Devant Le Risque: Critique Des Postulats Et Axiomes de l’École Américaine.” *Econometrica* 21 (4): 503–46.
- Anscombe, Francis J., and Robert J. Aumann. 1963. “A Definition of Subjective Probability.” *Annals of Mathematical Statistics* 34 (1): 199–205.
- Binz, Marcel, and Eric Schulz. 2023. “Using Cognitive Psychology to Understand GPT-3.” *Proceedings of the National Academy of Sciences* 120 (6): e2218523120.
- Camerer, Colin, and Martin Weber. 1992. “Recent Developments in Modeling Preferences: Uncertainty and Ambiguity.” *Journal of Risk and Uncertainty* 5 (4): 325–70.
- Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. “Stan: A Probabilistic Programming Language.” *Journal of Statistical Software* 76 (1): 1–32.
- Ellsberg, Daniel. 1961. “Risk, Ambiguity, and the Savage Axioms.” *The Quarterly Journal of Economics* 75 (4): 643–69.
- . 2001. *Risk, Ambiguity and Decision*. New York: Garland Publishing.
- Finetti, Bruno de. 1937. “Foresight: Its Logical Laws, Its Subjective Sources.” In *Studies in Subjective Probability*, edited by Henry E. Kyburg and Howard E. Smokler. Huntington, NY: Robert E. Krieger Publishing Co.
- Gelman, Andrew, Aki Vehtari, Daniel Simpson, Charles C. Margossian, Bob Carpenter, Yuling Yao, Lauren Kennedy, Jonah Gabry, Paul-Christian Bürkner, and Martin Modrák. 2020. “Bayesian Workflow.” *arXiv Preprint arXiv:2011.01808*.

- Ghirardato, Paolo, Fabio Maccheroni, and Massimo Marinacci. 2004. "Differentiating Ambiguity and Ambiguity Attitude." *Journal of Economic Theory* 118 (2): 133–73.
- Gilboa, Itzhak, and David Schmeidler. 1989. "Maxmin Expected Utility with Non-Unique Prior." *Journal of Mathematical Economics* 18 (2): 141–53.
- Hagendorff, Thilo, Sarah Fabi, and Michal Kosinski. 2023. "Human-Like Intuitive Behavior and Reasoning Biases Emerged in Large Language Models but Disappeared in ChatGPT." *Nature Computational Science* 3 (10): 833–38.
- Kahneman, Daniel, and Amos Tversky. 1979. "Prospect Theory: An Analysis of Decision Under Risk." *Econometrica* 47 (2): 263–91.
- Klibanoff, Peter, Massimo Marinacci, and Sujoy Mukerji. 2005. "A Smooth Model of Decision Making Under Ambiguity." *Econometrica* 73 (6): 1849–92.
- Knight, Frank H. 1921. *Risk, Uncertainty and Profit*. Boston: Houghton Mifflin.
- Levi, Isaac. 1980. *The Enterprise of Knowledge: An Essay on Knowledge, Credal Probability, and Chance*. Cambridge, MA: MIT Press.
- . 1986. *Hard Choices: Decision Making Under Unresolved Conflict*. Cambridge, UK: Cambridge University Press.
- Lieder, Falk, and Thomas L. Griffiths. 2020. "Resource-Rational Analysis: Understanding Human Cognition as the Optimal Use of Limited Computational Resources." *Behavioral and Brain Sciences* 43: e1.
- Luce, R. Duncan. 1959. "Individual Choice Behavior: A Theoretical Analysis."
- McFadden, Daniel. 1974. "Conditional Logit Analysis of Qualitative Choice Behavior." *Frontiers in Econometrics*, 105–42.
- Modrák, Martin, Angie H. Moon, Shinyoung Kim, Paul-Christian Bürkner, Niko Huurre, Kateřina Faltejsková, Andrew Gelman, and Aki Vehtari. 2023. "Simulation-Based Calibration Checking for Bayesian Computation: The Choice of Test Quantities Shapes Sensitivity." *Bayesian Analysis* 18 (2): 1–28.
- Neumann, John von, and Oskar Morgenstern. 1944. *Theory of Games and Economic Behavior*. Princeton, NJ: Princeton University Press.
- . 1947. *Theory of Games and Economic Behavior*. 2nd ed. Princeton, NJ: Princeton University Press.
- Ramsey, Frank P. 1926. "Truth and Probability." In *Studies in Subjective Probability*, edited by Henry E. Kyburg and Howard E. Smokler. Huntington, NY: Robert E. Krieger Publishing Co.
- Ruggeri, Kai, Sonia Alí, Mari Louise Berge, Giulia Bertoldo, Ludvig D. Bjørndal, Anna Cortijos-Bernabeu, Clair Davison, et al. 2020. "Replicating Patterns of Prospect Theory for Decision Under Risk." *Nature Human Behaviour* 4 (6): 622–33.
- Savage, Leonard J. 1954a. "The Foundations of Statistics."
- . 1954b. *The Foundations of Statistics*. New York: John Wiley & Sons.
- Simon, Herbert A. 1955. "A Behavioral Model of Rational Choice." *The Quarterly Journal of Economics* 69 (1): 99–118.
- Talts, Sean, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. 2018. "Validating Bayesian Inference Algorithms with Simulation-Based Calibration." *arXiv Preprint arXiv:1804.06788*.
- Thaler, Richard H., and Cass R. Sunstein. 2008. *Nudge: Improving Decisions about Health, Wealth, and Happiness*. New Haven, CT: Yale University Press.
- Train, Kenneth E. 2009. "Discrete Choice Methods with Simulation."
- Trautmann, Stefan T., and Gijs van de Kuilen. 2015. "Ambiguity Attitudes." In *The Wiley Blackwell Handbook of Judgment and Decision Making*, edited by Gideon Keren and George Wu, 89–116. Wiley-Blackwell.

- Tversky, Amos, and Daniel Kahneman. 1983. "Extensional Versus Intuitive Reasoning: The Conjunction Fallacy in Probability Judgment." *Psychological Review* 90 (4): 293–315.
- . 1992. "Advances in Prospect Theory: Cumulative Representation of Uncertainty." *Journal of Risk and Uncertainty* 5 (4): 297–323.